Building Cross-Domain Mapping Chains From Multi-CycleGAN for Hyperspectral Image Classification

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***Abstract*— The small-sample-size issue in hyperspectral image (HSI) classification remains a significant challenge. To improve the classification accuracy of a dataset with a few labeled samples (target domain), we can use knowledge from another dataset with sufficient labeled samples (source domain). This is called cross-domain HSI classification. Transfer learning enables knowledge transfer between source and target domains. Different HSI datasets are often acquired by different sensors, resulting in different characteristics. Consequently, different HSIs possess different feature spaces, and knowledge transfer between them becomes difficult due to heterogeneities. CycleGAN, based on adversarial learning, can help solve heterogeneous transfer learning tasks by establishing the two-way mapping between two different feature spaces. However, CycleGAN contains only one cycle for data, leading to large mapping errors. This article proposes a novel CycleGAN-based transfer learning method for cross-domain HSI classification. The proposed method extends the two-way mapping of CycleGAN. It incorporates multiple mapping cycles to construct a multi-CycleGAN, which is then unfolded to derive the Cross-Domain Mapping Chain (CDMC) model. The generators in our proposed CDMC provide accurate mappings between domains. Moreover, we calculate and accumulate errors in each cycle, and the backpropagation of accumulated errors through the chains improves the model's performance. Besides, auxiliary classifiers are introduced to account for class-conditional distributions in the mapping process. Experimental results on three real-world heterogeneous cross-domain HSI datasets show the effectiveness of the proposed method.**

***Index Terms*— Adversarial learning, cross-domain hyperspectral image (HSI) classification, Cross-Domain Mapping Chain (CDMC), heterogeneous transfer learning.**

I. Introduction

HYPERSPECTRAL image (HSI) classification is pivotal and foundational in many applications [1], [2], [3]. Deep learning has been widely used in hyperspectral remote sensing. With sufficient labeled samples for training, a large number of parameters in a deep learning-based model can be well-tuned to achieve impressive results [4]. Nevertheless, labeling HSI data is exceptionally time-consuming and labor-intensive, so most deep learning-based methods suffer from the small-sample-size problem [5], [6].

The perspective of many previous methods has been confined to a single HSI dataset. Fortunately, the advancement of transfer learning theory has expanded considerations to new domains. Transfer learning aims to leverage knowledge from the new domains [7], [8]. In the context of transfer learning, the source domain refers to a dataset with sufficient labeled samples in the transfer learning situation. In contrast, the target domain refers to a dataset with few labeled samples. Transfer learning reveals that the knowledge from the source domain can be used to improve the classification accuracy of the target domain [9]. However, in most cases, knowledge transfer becomes difficult because source and target domains have different spectral bands due to the differences in sensors. Such a task is called heterogeneous transfer learning, which constitutes the primary focus of this article.

Many cross-domain HSI classification methods based on transfer learning have been proposed in the last decade. Fine-tuning is a widely employed technique for knowledge transfer in HSI classification. In a fine-tuning-based approach, a specific network is trained first with sufficient labeled samples in the source domain. Afterward, the network is slightly adjusted, and then a small number of samples from the target domain are used to optimize the parameters. Some cross-domain HSI classification methods based on fine-tuning have been proposed, such as [10], [11], [12], [13], [14], [15], [16], [17]. However, fine-tuning may distort the pre-trained features in the source domain, resulting in poor generalization ability [18].

Feature space alignment is another prevalent transfer learning method for cross-domain HSI classification. The fundamental concept behind this method is that the probability distributions of data in the source and target domains typically differ in the initial feature space. However, mapping them into a shared feature space makes these distributions similar. Some cross-domain HSI classification methods based on feature space alignment have been proposed, such as [9], [19], [20], [21], [22], [23], [24], [25]. However, the task of learning a suitable feature space poses its own set of challenges. The extracted features in the shared space are commonly distorted by minimizing a metric that can indicate the discrepancy between different domains, such as maximum mean discrepancy (MMD). Feature distortion disrupts the intrinsic class-wise structures from source to target, thus reducing target classification performance [26].

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In many feature space alignment approaches, generative adversarial networks (GANs) are employed to facilitate learning a shared feature space. In the methods based on adversarial learning, a generator in GAN is utilized to output domain-invariant features of source and target samples. Concurrently, a discriminator undertakes the task of binary domain classification. The generator attempts to fool the discriminator so that it cannot correctly determine which domain a sample belongs to. By adversarial learning between the discriminator and the generator, the domain shift between source and target domains is reduced [27]. However, it is worth noting that most existing adversarial adaptation methods for remote sensing data do not consider class-conditional distributions [9].

Based on adversarial learning, CycleGAN [28] facilitates knowledge transfer between different domains through a two-way mapping process. In CycleGAN, source samples are initially mapped from the source domain to the target domain and reversed back to the source domain, forming a cycle. A similar cycle is applied to target samples, mapping from the target domain to the source domain and then back to the target domain. However, it is essential to note that CycleGAN conducts just a single cycle for the input data, which can result in substantial mapping errors and consequently impact the effectiveness of knowledge transfer. We aspire to enhance the two-way mapping between domains and make it more profitable for cross-domain HSI classification.

This article proposes a novel solution to address the challenge of heterogeneous cross-domain HSI classification, which we name Cross-Domain Mapping Chain (CDMC). Our approach draws inspiration from CycleGAN and establishes a two-way mapping between the source and target domains. The two-way mapping framework allows us to transfer knowledge between different domains without paired input–output instances. To enhance the mapping process, we introduce auxiliary classifiers that preserve discriminative information throughout the mapping procedure. These classifiers encourage the model to consider class-conditional distributions, facilitating its adaptation to the classification task. Subsequently, we apply multiple cycles of mappings to both source and target data, giving rise to the multicycle generative adversarial network (multi-CycleGAN). The multi-CycleGAN is unfolded through cycles later, and thus, two chains (i.e., a source chain and a target chain) for source and target data can be built. Unlike the original CycleGAN, which contains only one cycle for input data, our proposed CDMC has multiple cycles. Since the rich intermediate features produced by each cycle can be regarded as the original features that have been added with noise, the input data of the generators are further enhanced, improving the performance of the generators. The generators can learn more accurate mappings between different domains. In addition, errors are calculated and accumulated by each cycle, and the backpropagation through source and target chains improves the performance of our model. The main contributions of our work are stated as follows.

1) The proposed CDMC leverages a two-way mapping strategy to facilitate knowledge transfer between the source and target domains. Moreover, auxiliary classifiers are introduced to ensure the discriminability of samples in the mapping.
2) We conduct mappings between source and target domains through multiple cycles, creating the multi-CycleGAN. This multi-CycleGAN is unfolded to establish CDMC for both source and target data, allowing the generators within our model to learn precise mappings between these domains. Furthermore, the accumulated errors encourage our model to extract information from various cycles.
3) Our proposed CDMC can be used for heterogeneous HSIs. Experimental results on three cross-domain HSI datasets underscore the competitiveness and effectiveness of the proposed CDMC.

The remaining sections of this article are as follows. The related work to cross-domain classification, adversarial learning, and CycleGAN is briefly introduced in Section II. The proposed CDMC is described in Section III. The corresponding experiments and analyses are presented in Section IV. The conclusion is drawn in Section V.

II. Related Work

We present a brief review on cross-domain classification, adversarial learning, and CycleGAN in this section.

A. Cross-Domain Classification

In recent years, deep learning has provided opportunities to solve the HSI classification problem. However, the lack of labeled training samples in a single dataset may make it difficult to train the classifier. Fortunately, cross-domain classification methods [29], [30], [31] are proposed for HSI classification. In cross-domain classification methods, knowledge from the source domain can be used to assist in the classification of the target domain. Fine-tuning and feature space alignment are general methods for cross-domain HSI classification [32].

In a fine-tuning-based method, a network is fully pre-trained in the source domain, benefiting from an abundance of labeled samples to reach an optimal parameter state. Then, the network is transferred to the target domain for fine-tuning. Jiang et al. [10] propose a fine-tuning-based method for cross-domain HSI classification. In the model, 3-D convolutions are substituted by spatial and spectral separable 3-D convolutions so that fewer parameters are trained. The model above uses traditional dimensionality reduction methods so that source and target samples are transformed to have the same feature dimension. The dimensionality reduction is separate from the classification task. The method proposed by He et al. [11] leverages knowledge from ImageNet to enhance HSI classification. This method uses mapping layers to transform the target HSI to a dataset with three channels, which makes the target HSI lose much spectral information. Wang et al. [12] transfer the network pre-trained with mini-ImageNet to the spatial part of the HSI classification model. Zhang et al. [13] propose a method to solve the problem of cross-domain HSI classification based on fine-tuning. The network is much deeper and involves much smaller

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parameters than the conventional 3-D-convolutional neural network (CNN) for HSI classification. Gao et al. [14] combine fine-tuning and distillation strategies to propose a method for HSI classification. Knowledge from networks well-trained on source HSIs can be transferred to a distilled model. The feature representations are fine-tuned with small samples in the target HSIs. Zhong et al. [15] introduce a joint probability distribution adaptation approach to improve the effectiveness of knowledge transfer. Deng et al. [16] fine-tune the model by selecting a limited number of labeled samples using the corresponding active learning strategy. Su et al. [17] pre-train a graph convolutional network (GCN) using source data and then transfer it to target domain.

Feature space alignment aims to acquire a shared feature space in which source and target domain data exhibit similar probability distributions. In a study by Liu et al. [9], they present a network designed for cross-domain HSI classification capable of mapping source and target samples into a shared feature space. This approach employs the probability maximum mean discrepancy (PMMD) strategy to ensure that samples from distinct domains exhibit similar feature representations within the shared feature space. Zhang et al. [19] introduce graph optimal transport (GOT) to align topological information, which can facilitate MMD-based distribution alignment. In the method proposed by Li et al. [20], autoencoder networks are used to project source and target samples into a low-dimensional feature space. A multilevel MMD loss is imposed on the feature space to reduce the discrepancy between source and target domains. Wang et al. [21] make the distribution of the target domain match the class distribution in the source domain by MMD. Subsequently, the manifold regularization is utilized to keep the graph structure in the target domain. Zhen et al. [22] present a conditional MMD to achieve more comprehensive feature space alignment. In addition to aligning the local spatial information with MMD, Zhang et al. [23] introduce the GCN to align the topological structure of the two domains. Cheng et al. [24] combine meta-learning and causal learning to acquire transferable causal knowledge between source and target domains, thereby eliminating false statistical associations between class semantics and non-causal factors. Qu et al. [25] map source and target samples into a shared abundance space based on their physical characteristics and then design a classifier incorporating a 3-D-CNN for classification.

While fine-tuning and feature space alignment are frequently employed in cross-domain HSI classification, they present limitations. Fine-tuning-based methods may introduce distortions to pre-trained features from the source domain when applied in the target domain, potentially compromising the model's generalization abilities [18]. On the other hand, methods based on feature space alignment face the challenge of learning an appropriate feature space, as minimizing a metric to bridge the gap between domains may inadvertently distort the extracted features within the shared space [26].

B. Adversarial Learning

GANs have become widely adopted in diverse fields, such as object detection [33], [34], image classification [35], [36], target detection [37], and data synthesis [38], [39]. A typical GAN framework comprises a generator and a discriminator, which update parameters by adversarial learning. The generator takes noise as input and attempts to produce counterfeit samples indistinguishable from the real samples within a given dataset. The discriminator performs a binary classification task, determining whether the input samples are real or fake samples. Roy et al. [40] enhance the accuracy of HSI classification by utilizing the generated samples of a GAN. Feng et al. [41] propose a combination of self-supervised learning and adversarial learning as a solution for the HSI classification problem. Additionally, Feng et al. [42] introduce two GANs designed specifically for coarse-scale and fine-scale generation. They further enhance the sample quality of the generators by incorporating contrastive loss.

Adversarial learning can be introduced into transfer learning cases [43], [44]. Through adversarial learning between discriminators and generators, a feature space can be learned to achieve feature alignment for source and target domains. Several methods using adversarial learning to solve the problem of cross-domain HSI classification have been proposed. Li et al. [45] unify domain alignment and few-shot learning (FSL) for cross-domain HSI classification. The method uses adversarial learning to learn a shared feature space. Wang et al. [46] combine adversarial learning and broad learning to design a system for cross-domain HSI classification. Ma et al. [47] present a generator based on variational autoencoders, which can learn spatial–spectral features of different domains through adversarial learning. Yu et al. [48] develop a feature alignment method for cross-domain HSI classification, combining compact style and adversarial learning.

Although adversarial learning facilitates feature alignment between source and target domains, most available adversarial adaptation methods in remote sensing do not consider the class-conditional distributions [9].

C. CycleGAN

By employing adversarial learning, CycleGAN [28] is initially developed for image-to-image translation, which aims to learn a two-way mapping between different domains. The framework of CycleGAN is illustrated in Fig. 1. As can be seen in Fig. 1, to achieve the two-way mapping between source and target domains ($\mathcal{S}$ and $\mathcal{T}$, respectively), a common CycleGAN contains two generators: 1) a generator $G_{\mathcal{ST}}$ that performs the mapping $\mathcal{S} \rightarrow \mathcal{T}$ and 2) a generator $G_{\mathcal{TS}}$ that performs the mapping $\mathcal{T} \rightarrow \mathcal{S}$. Furthermore, there exist two discriminators: 1) a discriminator $D_{\mathcal{S}}$ that attempts to discriminate between the original and mapped source data and 2) a discriminator $D_{\mathcal{T}}$ that attempts to discriminate between the original and mapped target data. The adversarial training between generators and discriminators can help in learning two-way mappings. In addition, the mapping is expected to be cycle-consistent, and it ensures that the mapped data can be re-translated to their original domain, i.e., $G_{\mathcal{ST}}(G_{\mathcal{TS}}(\mathbf{x}_{\mathcal{T}})) \approx \mathbf{x}_{\mathcal{T}}$ and $G_{\mathcal{TS}}(G_{\mathcal{ST}}(\mathbf{x}_{\mathcal{S}})) \approx \mathbf{x}_{\mathcal{S}}$. The adversarial loss encourages knowledge transfer between domains that only consider the marginal distributions. Thus, the two-way mapping can

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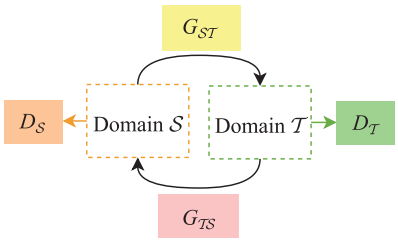

Fig. 1. Framework of CycleGAN.

work without paired training instances $\{\mathbf{x}_{\mathcal{S}}^i, \mathbf{x}_{\mathcal{T}}^i\}_{i=1}^N$, where a correspondence exists between the source sample $\mathbf{x}_{\mathcal{S}}^i$ and the target sample $\mathbf{x}_{\mathcal{T}}^i$.

CycleGAN can be used for cross-domain image classification. In order to improve the classification accuracy of remote sensing images, Zheng et al. [49] utilize CycleGAN to transform the training data of domain $\mathcal{S}$ into data with characteristics of domain $\mathcal{T}$, which reduces the workload of manually labeling data. Wang et al. [50] utilize CycleGAN to capture the transferable features between domains $\mathcal{S}$ and $\mathcal{T}$ for HSI classification.

Through the two-way mapping, knowledge can be translated between domains $\mathcal{S}$ and $\mathcal{T}$. Nevertheless, one limitation of CycleGAN is its restriction to performing mapping operations for input data within a single cycle, ultimately resulting in suboptimal generator performance. Poor generators lead to large mapping errors, affecting classification performance. To enhance the performance of the generators for image synthesis, Liu et al. [51] propose a method that employs a pseudo-cycle consistent module to add half a cycle for data. In contrast to the original CycleGAN, the generators in the method proposed by Liu et al. [51] exhibit improved robustness and performance. According to this idea, this article extends the single cycle of mapping to multiple cycles.

III. Proposed Method

The heterogeneous cross-domain HSI classification task contains two domains. The source domain is represented by $\mathcal{S} = \{\mathbf{x}_{\mathcal{S}}^i, \mathrm{y}_{\mathcal{S}}^i\}_{i=1}^{N_{\mathcal{S},\mathrm{all}}}$. $\mathbf{x}_{\mathcal{S}}^i \in \mathbb{R}^{P\times P\times B_{\mathcal{S}}}$ is the ith sample patch in the source domain, where P represents the patch size, $B_{\mathcal{S}}$ represents the number of spectral bands, and $\mathrm{y}_{\mathcal{S}}^i$ is the corresponding land-cover label of $\mathbf{x}_{\mathcal{S}}^i$. Similarly, the target domain is represented by $\mathcal{T} = \{\mathbf{x}_{\mathcal{T}}^i, \mathrm{y}_{\mathcal{T}}^i\}_{i=1}^{N_{\mathcal{T},\mathrm{all}}}$. $\mathbf{x}_{\mathcal{T}}^i \in \mathbb{R}^{P\times P\times B_{\mathcal{T}}}$ is the ith sample patch in the target domain, where $B_{\mathcal{T}}$ represents the number of spectral bands, and $\mathrm{y}_{\mathcal{T}}^i$ is the corresponding land-cover label of $\mathbf{x}_{\mathcal{T}}^i$. We should note that the domains $\mathcal{S}$ and $\mathcal{T}$ share the same C land-cover classes. Additionally, in the heterogeneous cross-domain HSI classification task, the domains $\mathcal{S}$ and $\mathcal{T}$ have different spectral bands (i.e., $B_{\mathcal{S}} \neq B_{\mathcal{T}}$), so knowledge transfer becomes difficult.

This section describes the proposed method named CDMC in detail. The CDMC can be applicable to heterogeneous cross-domain HSI classification. Our work achieves knowledge transfer through a two-way mapping. Class-conditional distributions are considered thanks to the added auxiliary classifiers. We conduct the mappings between domains $\mathcal{S}$ and $\mathcal{T}$ in multiple cycles to form the multi-CycleGAN. Unfolding the multi-CycleGAN, CDMC can be obtained.

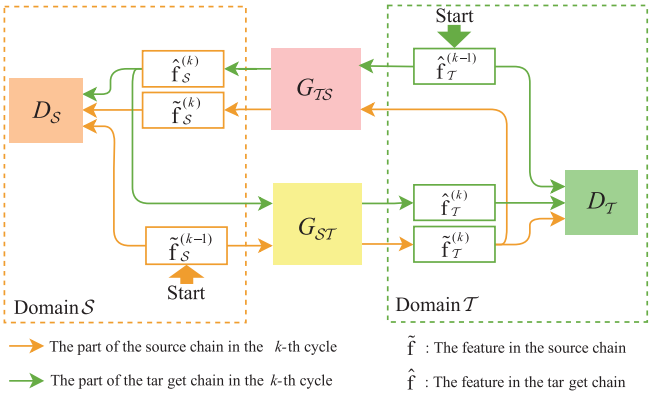

Fig. 2. Framework of the kth cycle in our proposed CDMC. $G_{\mathcal{ST}}$ is a generator that performs the mapping $\mathcal{S} \to \mathcal{T}$, and $G_{\mathcal{TS}}$ is a generator that performs the mapping $\mathcal{T} \to \mathcal{S}$. $D_{\mathcal{T}}$ is a discriminator in the domain $\mathcal{T}$, and $D_{\mathcal{S}}$ is a discriminator in the domain $\mathcal{S}$.

A. Auxiliary Classifiers for Class-Conditional Distributions

Fig. 2 depicts the two-way mapping process within our proposed CDMC during the kth cycle. Our proposed CDMC contains two chains: a source chain and a target chain. The source chain serves as the mapping path for any source sample, while the target chain serves as the mapping path for any target sample. Each cycle contains a part of the source chain and a part of the target chain, correspondingly. $G_{\mathcal{ST}}$ and $G_{\mathcal{TS}}$ are shared in the source and target chains. We use $\tilde{\mathbf{f}} \in \mathbb{R}^L$ to represent the features in the source chain and $\hat{\mathbf{f}} \in \mathbb{R}^L$ to represent the features in the target chain, where L is the feature dimension. As can be seen in Fig. 2, through the two-way mapping, the feature in the part of the source chain undergoes from the domain $\mathcal{S}$ to the domain $\mathcal{T}$ and back to the domain $\mathcal{S}$ ($\tilde{\mathbf{f}}_{\mathcal{S}}^{(k-1)} \to \tilde{\mathbf{f}}_{\mathcal{T}}^{(k)} \to \tilde{\mathbf{f}}_{\mathcal{S}}^{(k)}$), and the feature in the part the target chain undergoes from the domain $\mathcal{T}$ to the domain $\mathcal{S}$ and back to the domain $\mathcal{T}$ ($\hat{\mathbf{f}}_{\mathcal{T}}^{(k-1)} \to \hat{\mathbf{f}}_{\mathcal{S}}^{(k)} \to \hat{\mathbf{f}}_{\mathcal{T}}^{(k)}$), so it is called the cycle in this article. Notice that $k \in \{1, 2, 3, \ldots, K\}$, and K is the number of cycles. In addition, we use $\tilde{\mathbf{f}}_{\mathcal{S}}^{(0)}$ and $\hat{\mathbf{f}}_{\mathcal{T}}^{(0)}$ to represent the original source and target features before mapping, respectively.

By leveraging adversarial learning, the two-way mapping facilitates knowledge transfer between domains $\mathcal{S}$ and $\mathcal{T}$. However, in cross-domain classification tasks, it becomes imperative to account for class-conditional distributions [9]. The mapping should ensure that the class labels are consistent. The mapped features can be classified to the same class as before mapping. That is, for the ith source sample, the classifier should assign the same class label to the features $\tilde{\mathbf{f}}_{\mathcal{S}}^{(k-1),i}$, $\tilde{\mathbf{f}}_{\mathcal{T}}^{(k),i}$ and $\tilde{\mathbf{f}}_{\mathcal{S}}^{(k),i}$ in the kth cycle. Similarly, for the ith target sample, the classifier should assign the same class label to the features $\hat{\mathbf{f}}_{\mathcal{T}}^{(k-1),i}$, $\hat{\mathbf{f}}_{\mathcal{S}}^{(k),i}$, and $\hat{\mathbf{f}}_{\mathcal{T}}^{(k),i}$ in the kth cycle.

Thus, auxiliary classifiers are added to each discriminator [52]. For the discriminator in the domain $\mathcal{T}$ ($D_{\mathcal{T}}$), it contains the following two classifiers.

1) A domain classifier ($D_{\mathcal{T},\mathrm{dom}}$ in Fig. 3) performs a domain classification task. Specifically, $D_{\mathcal{T},\mathrm{dom}}$ distinguishes whether the input features are original or mapped target features.
2) An auxiliary land-cover classifier ($D_{\mathcal{T},\mathrm{cls}}$ in Fig. 3) performs a land-cover classification task in the domain $\mathcal{T}$.

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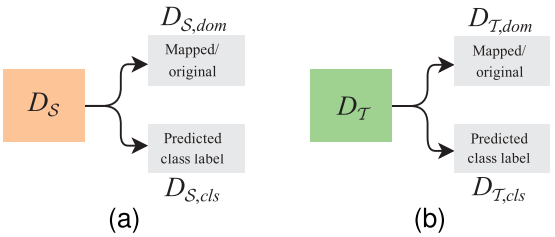

Fig. 3. Two classifiers contained in the discriminators $D_{\mathcal{S}}$ and $D_{\mathcal{T}}$. (a) $D_{\mathcal{S}}$. (b) $D_{\mathcal{T}}$. Auxiliary classifiers ($D_{\mathcal{S},\text{cls}}$ and $D_{\mathcal{T},\text{cls}}$) are added to account for the class-conditional distribution, making the model more suitable for classification tasks.

The land-cover classifier contributes to the alignment of class-conditional distributions across mappings, which also helps to improve classification accuracy.

Similar to $D_{\mathcal{T}}$, the discriminator in the domain $\mathcal{S}$ ($D_{\mathcal{S}}$) contains two classifiers: 1) a domain classifier ($D_{\mathcal{S},dom}$) and 2) an auxiliary land-cover classifier ($D_{\mathcal{S},\text{cls}}$). With the added auxiliary classifiers, class-conditional distributions in the mapping process can be maintained.

B. Multi-CycleGAN and CDMC

Through the added auxiliary classifiers, the two-way mapping considers the class-conditional distribution between domains $\mathcal{S}$ and $\mathcal{T}$. Then, we perform mappings between different domains in multiple cycles to form multi-CycleGAN. The intermediate source and target features produced by each cycle are considered as the input for the next cycle. Specifically, features $\widetilde{\mathbf{f}}_{\mathcal{S}}^{(k)}$ and $\hat{\mathbf{f}}_{\mathcal{T}}^{(k)}$ (where $k \in \{1, 2, 3, \ldots, K-1\}$) are fed into the generators $G_{\mathcal{ST}}$ and $G_{\mathcal{TS}}$ to start the next cycle. It is worth noting that the generators and discriminators are shared in each cycle. The original CycleGAN performs only one cycle for input data, while our method performs multiple cycles across domains. The intermediate features $\widetilde{\mathbf{f}}_{\mathcal{S}}^{(k)}$ and $\hat{\mathbf{f}}_{\mathcal{T}}^{(k)}$ can be regarded as the reconstructions of the original features $\widetilde{\mathbf{f}}_{\mathcal{S}}^{(0)}$ and $\hat{\mathbf{f}}_{\mathcal{T}}^{(0)}$, i.e., $\widetilde{\mathbf{f}}_{\mathcal{S}}^{(k)} \approx \widetilde{\mathbf{f}}_{\mathcal{S}}^{(0)}$ and $\hat{\mathbf{f}}_{\mathcal{T}}^{(k)} \approx \hat{\mathbf{f}}_{\mathcal{T}}^{(0)}$, and this operation acts as adding noise to the original features. The input data of the generators are further augmented, improving the performance of the generators. In essence, multiple cycles for input data afford the model increased opportunities to rectify potential errors that may arise during the mapping process. As a result, the generators within our model can learn more precise feature mappings. Simply adding noise to features for data augmentation may have a negative impact on the classification task. Such a method could lead to processed features that no longer resemble the same class as the original or even any known classes [53]. On the contrary, with the help of auxiliary classifiers, our generators guarantee the consistency of classes before and after mapping.

Then, multi-CycleGAN is unfolded through cycles, and CDMC can be obtained. Fig. 4 illustrates the target chain unfolded through cycles. The proposed model also contains another source chain. To view the framework of the source chain, we only need to swap $\mathcal{T}$ and $\mathcal{S}$ and replace $\hat{\mathbf{f}}$ with $\widetilde{\mathbf{f}}$ in Fig. 4. $E_{\mathcal{T}}$ is a spatial–spectral feature extractor in the domain $\mathcal{T}$, consisting of 2-D and 3-D convolution layers. Meanwhile, the generators and discriminators in our work are composed of fully connected layers. The generators are shared not only within each cycle but also across the source and target chains, thereby effectively reducing the overall number of parameters in our proposed model. Furthermore, errors are accumulated over each cycle, and our CDMC framework considers knowledge from all cycles within the chains. Fig. 5 illustrates the progression from Target-only to Cycle-AC-GAN to CDMC. In Fig. 5, the Target-only method trains the model using a small number of target samples. The Cycle-AC-GAN method performs mappings for source and target data in a single cycle ($K = 1$). Compared to Target-only and Cycle-AC-GAN, our proposed CDMC forms source and target chains of mappings between domains $\mathcal{S}$ and $\mathcal{T}$. In the proposed CDMC, the intermediate features in long source and target chains enrich the representations of samples. The backpropagation of the accumulated errors through the chains helps to improve the classification performance of the model.

For the original feature of a source sample $\widetilde{\mathbf{f}}_{\mathcal{S}}^{(0)}$, there exists the following mapping process in our proposed CDMC's source chain: $\widetilde{\mathbf{f}}_{\mathcal{S}}^{(0)} \to \widetilde{\mathbf{f}}_{\mathcal{T}}^{(1)} \to \widetilde{\mathbf{f}}_{\mathcal{S}}^{(1)} (\approx \widetilde{\mathbf{f}}_{\mathcal{S}}^{(0)}) \to \widetilde{\mathbf{f}}_{\mathcal{T}}^{(2)} \to \ldots \to \widetilde{\mathbf{f}}_{\mathcal{T}}^{(K)} \to \widetilde{\mathbf{f}}_{\mathcal{S}}^{(K)} (\approx \widetilde{\mathbf{f}}_{\mathcal{S}}^{(0)})$. Similarly, for the original feature of a target sample $\hat{\mathbf{f}}_{\mathcal{T}}^{(0)}$, there exists the mapping process in the target chain: $\hat{\mathbf{f}}_{\mathcal{T}}^{(0)} \to \hat{\mathbf{f}}_{\mathcal{S}}^{(1)} \to \hat{\mathbf{f}}_{\mathcal{T}}^{(1)} (\approx \hat{\mathbf{f}}_{\mathcal{T}}^{(0)}) \to \hat{\mathbf{f}}_{\mathcal{S}}^{(2)} \to \ldots \to \hat{\mathbf{f}}_{\mathcal{S}}^{(K)} \to \hat{\mathbf{f}}_{\mathcal{T}}^{(K)} (\approx \hat{\mathbf{f}}_{\mathcal{T}}^{(0)})$.

For brevity, the original feature of a source sample $\mathbf{x}_{\mathcal{S}}$ can be obtained by

$$\widetilde{\mathbf{f}}_{\mathcal{S}}^{(0)} = \text{Flatten}(E_{\mathcal{S}}(\mathbf{x}_{\mathcal{S}})) \tag{1}$$

where Flatten$(\cdot)$ represents the operation to flatten the feature. The mapping process of the source chain in the kth cycle can be described as follows:

$$\widetilde{\mathbf{f}}_{\mathcal{T}}^{(k)} = G_{\mathcal{ST}}\left(\text{Concat}\left(\widetilde{\mathbf{f}}_{\mathcal{S}}^{(k-1)}, \mathbf{y}_{\mathcal{S},\text{one-hot}}, \mathbf{z}\right)\right) \tag{2}$$

$$\widetilde{\mathbf{f}}_{\mathcal{S}}^{(k)} = G_{\mathcal{TS}}\left(\text{Concat}\left(\widetilde{\mathbf{f}}_{\mathcal{T}}^{(k)}, \mathbf{y}_{\mathcal{S},\text{one-hot}}, \mathbf{z}\right)\right) \tag{3}$$

where $\mathbf{z} \in \mathbb{R}^d$ is a random noise vector sampled from $\mathcal{N}(0, \sigma^2)$. Concat$(\cdot)$ represents the operation to concatenate, and $\mathbf{y}_{\mathcal{S},\text{one-hot}} \in \mathbb{R}^C$ represents the one-hot encoding for the class label of $\mathbf{x}_{\mathcal{S}}$. Similarly, the original feature of a target sample $\mathbf{x}_{\mathcal{T}}$ can be obtained by

$$\hat{\mathbf{f}}_{\mathcal{T}}^{(0)} = \text{Flatten}(E_{\mathcal{T}}(\mathbf{x}_{\mathcal{T}})). \tag{4}$$

The mapping process of the target chain in the kth cycle can be described as follows:

$$\hat{\mathbf{f}}_{\mathcal{S}}^{(k)} = G_{\mathcal{TS}}\left(\text{Concat}\left(\hat{\mathbf{f}}_{\mathcal{T}}^{(k-1)}, \mathbf{y}_{\mathcal{T},\text{one-hot}}, \mathbf{z}\right)\right) \tag{5}$$

$$\hat{\mathbf{f}}_{\mathcal{T}}^{(k)} = G_{\mathcal{ST}}\left(\text{Concat}\left(\hat{\mathbf{f}}_{\mathcal{S}}^{(k)}, \mathbf{y}_{\mathcal{T},\text{one-hot}}, \mathbf{z}\right)\right). \tag{6}$$

We calculate the loss function in each cycle. The loss function of discriminators and spatial–spectral feature extractors involves adversarial loss and classification loss in the kth cycle

$$\mathcal{L}_{D,E}^{k} = \mathcal{L}_{D,E,\text{adv}}^{k} + \alpha \mathcal{L}_{D,E,\text{cls}}^{k} \tag{7}$$

where α is a balancing parameter, $\mathcal{L}_{D,\text{adv}}^{k}$ represents the adversarial loss in the kth cycle, and $\mathcal{L}_{D,\text{cls}}^{k}$ represents the

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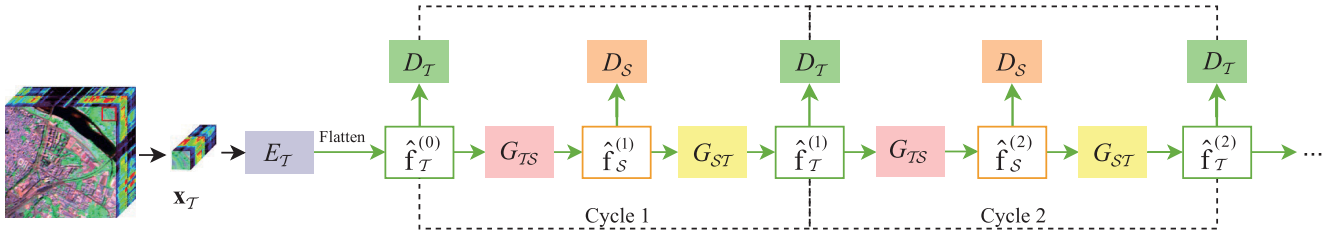

Fig. 4. Target chain unfolded through cycles in our proposed CDMC. The features generated by each cycle are fed into the next cycle. Note that the proposed CDMC also includes a source chain. To obtain a view of the source chain, swap $\mathcal{T}$ and $\mathcal{S}$, and replace $\hat{\mathbf{f}}$ with $\widetilde{\mathbf{f}}$.

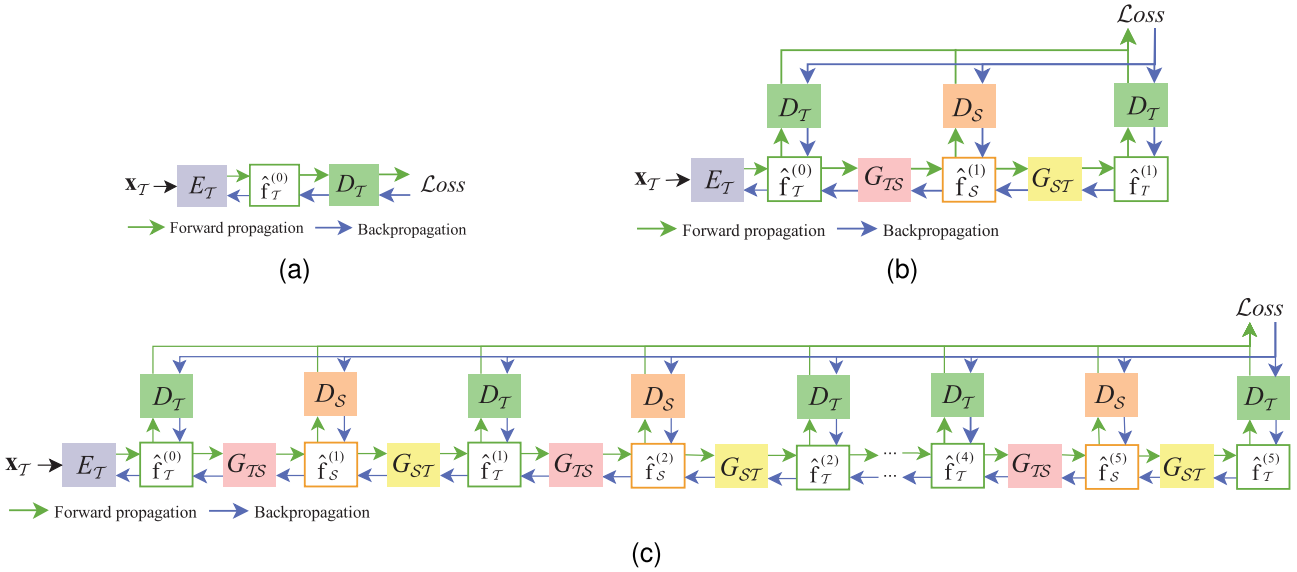

Fig. 5. Backpropagation of different methods. (a) Target-only. (b) Cycle-AC-GAN ($K = 1$). (c) CDMC ($K = 5$). The figure only shows the target parts of Cycle-AC-GAN and CDMC, and their source parts are not plotted. The feature representations are enriched by the long chains in our proposed CDMC, and the backpropagation of the accumulated errors through the chains facilitates the model to improve the classification performance.

classification loss in the kth cycle. The $\mathcal{L}^k_{D,\text{adv}}$ is described as follows:

$$\begin{aligned}\mathcal{L}^k_{D,E,\text{adv}} = \max_{\theta_D,\theta_E} \frac{1}{N_\mathcal{T}} \sum_{i=1}^{N_\mathcal{T}} \Big[&\log\Big(1 - D_{\mathcal{S},\text{dom}}\Big(\hat{\mathbf{f}}^{(k),i}_\mathcal{S}\Big)\Big) \\ &+ \log\Big(1 - D_{\mathcal{T},\text{dom}}\Big(\hat{\mathbf{f}}^{(k),i}_\mathcal{T}\Big)\Big)\Big] \\ + \frac{1}{N_\mathcal{S}} \sum_{i=1}^{N_\mathcal{S}} \Big[&\log\Big(1 - D_{\mathcal{T},\text{dom}}\Big(\widetilde{\mathbf{f}}^{(k),i}_\mathcal{T}\Big)\Big) \\ &+ \log\Big(1 - D_{\mathcal{S},\text{dom}}\Big(\widetilde{\mathbf{f}}^{(k),i}_\mathcal{S}\Big)\Big)\Big] \end{aligned} \tag{8}$$

where $N_\mathcal{S}$ and $N_\mathcal{T}$ refer to the number of source and target samples for training, respectively. θ_D and θ_E refer to the trainable parameters in discriminators and feature extractors, respectively. The $\mathcal{L}^k_{D,\text{cls}}$ is described as

$$\begin{aligned}&\mathcal{L}^k_{D,E,\text{cls}} \\ &= \max_{\theta_D,\theta_E} \frac{1}{N_\mathcal{T}} \sum_{i=1}^{N_\mathcal{T}} \Big[-\mathcal{L}^{\hat{\mathbf{f}}^{(k),i}_\mathcal{S}}_{\text{ce}} - \mathcal{L}^{\hat{\mathbf{f}}^{(k),i}_\mathcal{T}}_{\text{ce}}\Big] + \frac{1}{N_\mathcal{S}} \sum_{i=1}^{N_\mathcal{S}} \Big[-\mathcal{L}^{\widetilde{\mathbf{f}}^{(k),i}_\mathcal{T}}_{\text{ce}} - \mathcal{L}^{\widetilde{\mathbf{f}}^{(k),i}_\mathcal{S}}_{\text{ce}}\Big]. \end{aligned} \tag{9}$$

In (9), $\mathcal{L}^{\widetilde{\mathbf{f}}^{(k),i}_\mathcal{S}}_{\text{ce}}$ and $\mathcal{L}^{\hat{\mathbf{f}}^{(k),i}_\mathcal{S}}_{\text{ce}}$ are the cross-entropy losses produced by $D_{\mathcal{S},\text{cls}}$ predicting the class label of $\widetilde{\mathbf{f}}^{(k),i}_\mathcal{S}$ and $\hat{\mathbf{f}}^{(k),i}_\mathcal{S}$, respectively. $\mathcal{L}^{\hat{\mathbf{f}}^{(k),i}_\mathcal{T}}_{\text{ce}}$ and $\mathcal{L}^{\widetilde{\mathbf{f}}^{(k),i}_\mathcal{T}}_{\text{ce}}$ are the cross-entropy losses produced by $D_{\mathcal{T},\text{cls}}$ predicting the class label of $\hat{\mathbf{f}}^{(k),i}_\mathcal{T}$ and $\widetilde{\mathbf{f}}^{(k),i}_\mathcal{T}$, respectively.

The input to each generator is a concatenation containing a sample's feature, the one-hot encoding of the class label, and the random noise vector. In each cycle, the reconstructions of the original source and target features can be obtained, i.e., $\widetilde{\mathbf{f}}^{(k),i}_\mathcal{S} \approx \widetilde{\mathbf{f}}^{(0),i}_\mathcal{S}$ and $\hat{\mathbf{f}}^{(k),i}_\mathcal{T} \approx \hat{\mathbf{f}}^{(0),i}_\mathcal{T}$. Cycle consistency should be kept in mappings [28]. In order to strengthen the cycle consistency, the loss function of generators in the kth cycle should include reconstruction loss, in addition to adversarial loss and classification loss

$$\mathcal{L}^k_G = \mathcal{L}^k_{G,\text{adv}} + \beta \mathcal{L}^k_{G,\text{cls}} + \lambda \mathcal{L}^k_{\text{rec}} \tag{10}$$

where β and λ are balancing parameters. To achieve the two-way mapping, the generators attempt to fool the discriminators, and the adversarial loss $\mathcal{L}^k_{G,\text{adv}}$ is described as

$$\begin{aligned}\mathcal{L}^k_{G,\text{adv}} = \min_{\theta_G} \frac{1}{N_\mathcal{T}} \sum_{i=1}^{N_\mathcal{T}} \Big[&\log\Big(1 - D_{\mathcal{S},\text{dom}}\Big(\hat{\mathbf{f}}^{(k),i}_\mathcal{S}\Big)\Big) \\ &+ \log\Big(1 - D_{\mathcal{T},\text{dom}}\Big(\hat{\mathbf{f}}^{(k),i}_\mathcal{T}\Big)\Big)\Big] \\ + \frac{1}{N_\mathcal{S}} \sum_{i=1}^{N_\mathcal{S}} \Big[&\log\Big(1 - D_{\mathcal{T},\text{dom}}\Big(\widetilde{\mathbf{f}}^{(k),i}_\mathcal{T}\Big)\Big) \\ &+ \log\Big(1 - D_{\mathcal{S},\text{dom}}\Big(\widetilde{\mathbf{f}}^{(k),i}_\mathcal{S}\Big)\Big)\Big] \end{aligned} \tag{11}$$

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where θ_G refers to the trainable parameters in the generators. The classification loss $\mathcal{L}^k_{G,\text{cls}}$ is as follows:

$$\mathcal{L}^k_{G,\text{cls}} = \min_{\theta_G} \frac{1}{N_\mathcal{T}} \sum_{i=1}^{N_\mathcal{T}} \left[\mathcal{L}_{\text{ce}}^{\hat{\mathbf{f}}_\mathcal{S}^{(k),i}} + \mathcal{L}_{\text{ce}}^{\hat{\mathbf{f}}_\mathcal{T}^{(k),i}} \right] + \frac{1}{N_\mathcal{S}} \sum_{i=1}^{N_\mathcal{S}} \left[\mathcal{L}_{\text{ce}}^{\widetilde{\mathbf{f}}_\mathcal{T}^{(k),i}} + \mathcal{L}_{\text{ce}}^{\widetilde{\mathbf{f}}_\mathcal{S}^{(k),i}} \right]. \quad (12)$$

$\mathcal{L}^k_{G,\text{cls}}$ allows generators to consider class-conditional distributions in the mapping. The reconstruction loss $\mathcal{L}^k_{\text{rec}}$ is defined as

$$\mathcal{L}^k_{\text{rec}} = \min_{\theta_G} \frac{1}{N_\mathcal{T}} \sum_{i=1}^{N_\mathcal{T}} \frac{1}{d_\mathcal{T}} \|\hat{\mathbf{f}}_\mathcal{T}^{(k),i} - \hat{\mathbf{f}}_\mathcal{T}^{(0),i}\|_2^2 + \frac{1}{N_\mathcal{S}} \sum_{i=1}^{N_\mathcal{S}} \frac{1}{d_\mathcal{S}} \|\widetilde{\mathbf{f}}_\mathcal{S}^{(k),i} - \widetilde{\mathbf{f}}_\mathcal{S}^{(0),i}\|_2^2 \quad (13)$$

where $d_\mathcal{S}$ and $d_\mathcal{T}$ are the dimensions of each source and target feature, respectively. $\|\cdot\|_2$ represents the Euclidean norm. $\mathcal{L}^k_{\text{rec}}$ ensures the cycle consistency.

The overall loss function of discriminators and feature extractors is accumulated by the losses of original features and mapped features in each cycle

$$\mathcal{L}^{\text{Total}}_{D,E} = \mathcal{L}_{\mathbf{f}^{(0)}} + \sum_{k=1}^{K} \gamma_k \mathcal{L}^k_{D,E} \quad (14)$$

where γ_k is the weight of the kth cycle. $\mathcal{L}_{\mathbf{f}^{(0)}}$ is the loss produced by the original source and target features before mapping, and it contains the domain classification loss and the cross-entropy loss

$$\mathcal{L}_{\mathbf{f}^{(0)}} = \max_{\theta_D,\theta_E} \frac{1}{N_\mathcal{T}} \sum_{i=1}^{N_\mathcal{T}} \left[\log\left(D_{\mathcal{T},\text{dom}}\left(\hat{\mathbf{f}}_\mathcal{T}^{(0),i} \right) \right) - \mathcal{L}_{\text{ce}}^{\hat{\mathbf{f}}_\mathcal{T}^{(0),i}} \right] + \frac{1}{N_\mathcal{S}} \sum_{i=1}^{N_\mathcal{S}} \left[\log\left(D_{\mathcal{S},\text{dom}}\left(\widetilde{\mathbf{f}}_\mathcal{S}^{(0),i} \right) \right) - \mathcal{L}_{\text{ce}}^{\widetilde{\mathbf{f}}_\mathcal{S}^{(0),i}} \right]. \quad (15)$$

The overall loss function of generators is accumulated by the loss function of generators in each cycle

$$\mathcal{L}^{\text{Total}}_{G} = \sum_{k=1}^{K} \gamma_k \mathcal{L}^k_{G}. \quad (16)$$

The proposed CDMC is summarized in Algorithm 1.

Algorithm 1 CDMC

Input:

Source samples with labels $\{\mathbf{x}^i_\mathcal{S}, \text{y}^i_\mathcal{S}\}_{i=1}^{N_\mathcal{S}}$, a small number of target samples with labels: $\{\mathbf{x}^i_\mathcal{T}, \text{y}^i_\mathcal{T}\}_{i=1}^{N_\mathcal{T}}$, the number of cycles K, the number of training epoch N, hyperparameters α, β and λ.

Initialize $\{\theta_{E_\mathcal{S}}, \theta_{E_\mathcal{T}}, \theta_{G_{\mathcal{ST}}}, \theta_{G_{\mathcal{TS}}}, \theta_{D_\mathcal{S}}$ and $\theta_{D_\mathcal{T}}\}$.
for $n = 1 : N$ **do**
  Calculate $\{\widetilde{\mathbf{f}}_\mathcal{S}^{(0),i}\}_{i=1}^{N_\mathcal{S}}$ using Eq. (1).
  Calculate $\{\hat{\mathbf{f}}_\mathcal{T}^{(0),i}\}_{i=1}^{N_\mathcal{T}}$ using Eq. (4).
  for $k = 1 : K$ **do**
    Calculate $\{\widetilde{\mathbf{f}}_\mathcal{T}^{(k),i}, \widetilde{\mathbf{f}}_\mathcal{S}^{(k),i}\}_{i=1}^{N_\mathcal{S}}$ using Eqs. (2)-(3).
    Calculate $\{\hat{\mathbf{f}}_\mathcal{S}^{(k),i}, \hat{\mathbf{f}}_\mathcal{T}^{(k),i}\}_{i=1}^{N_\mathcal{T}}$ using Eqs. (5)-(6).
  end for
  Update θ_D and θ_E by maximizing the adversarial loss using Eq. (14).
  Update θ_G by minimizing the adversarial loss using Eq. (16).
end for
return $\{\theta_{E_\mathcal{S}}, \theta_{E_\mathcal{T}}, \theta_{G_{\mathcal{ST}}}, \theta_{G_{\mathcal{TS}}}, \theta_{D_\mathcal{S}}$ and $\theta_{D_\mathcal{T}}\}$.

IV. Experiments

A. Data Description

In order to verify the effectiveness of the proposed CDMC, the experiments are conducted on the following three heterogeneous cross-domain HSI datasets.

1) **EHangzhou-RPaviaHR** dataset. This cross-domain HSI dataset comprises a source domain (EHangzhou dataset) and a target domain (RPaviaHR dataset). The EHangzhou dataset is derived from EO-1 Hyperion hyperspectral sensor imaging conducted over Hangzhou, China, while it contains 590 × 230 pixels and 198 spectral bands. The RPaviaHR dataset is obtained by ROSIS HSI sensor imaging over Pavia, Italy, while it contains 1400 × 512 pixels and 102 spectral bands. This pair of HSIs shares three land-cover classes. The false-color images and ground-truth maps of the EHangzhou-RPaviaHR dataset can be seen in Fig. 6.
2) **RPaviaU-DPaviaC** dataset. This cross-domain HSI dataset contains a source domain (RPaviaU dataset) and a target domain (DPaviaC dataset). The RPaviaU dataset is obtained by ROSIS HSI sensor imaging over the University of Pavia, while it contains 610 × 340 pixels and 103 spectral bands. The DPaviaC dataset is obtained by DAIS sensor imaging over Pavia, while it contains 400 × 400 pixels and 72 spectral bands. This pair of HSIs shares seven land-cover classes. The false-color images and ground-truth maps of the RPaviaU-DPaviaC dataset can be seen in Fig. 7.
3) **Houston2018–Houston2013** dataset. In this cross-domain HSI dataset, the Houston2018 dataset [54] is used as the source domain, and the Houston2013 dataset [55] is used as the target domain. The Houston2018 dataset contains 601 × 2384 pixels and 48 bands, while the Houston2013 dataset contains 349 × 1905 pixels and 144 bands. Seven common land-cover classes are selected from two datasets for cross-domain HSI classification. The false-color images and ground-truth maps of the Houston2018–Houston2013 dataset can be seen in Fig. 8.

The number of each land-cover class in EHangzhou-RPaviaHR, RPaviaU-DPaviaC and Houston2018–Houston2013 datasets is shown in Table I.

B. Experimental Settings

All experiments in this article are conducted on a machine with an Intel Core i5-11260H CPU, a GeForce GTX 3060 GPU, and 16 GB RAM. Moreover, PyTorch is used

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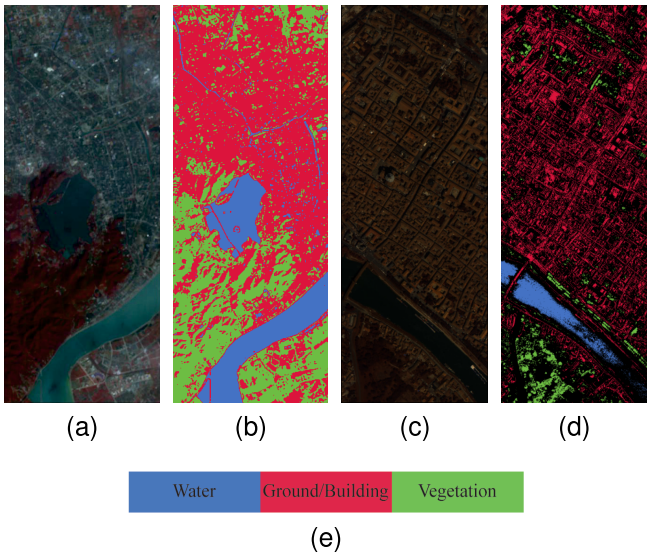

Fig. 6. False-color images and ground-truth maps of EHangzhou-RPaviaHR dataset. (a) False-color image of EHangzhou dataset. (b) Ground-truth map of EHangzhou dataset. (c) False-color image of RPaviaHR dataset. (d) Ground-truth map of RPaviaHR dataset. (e) Class legend.

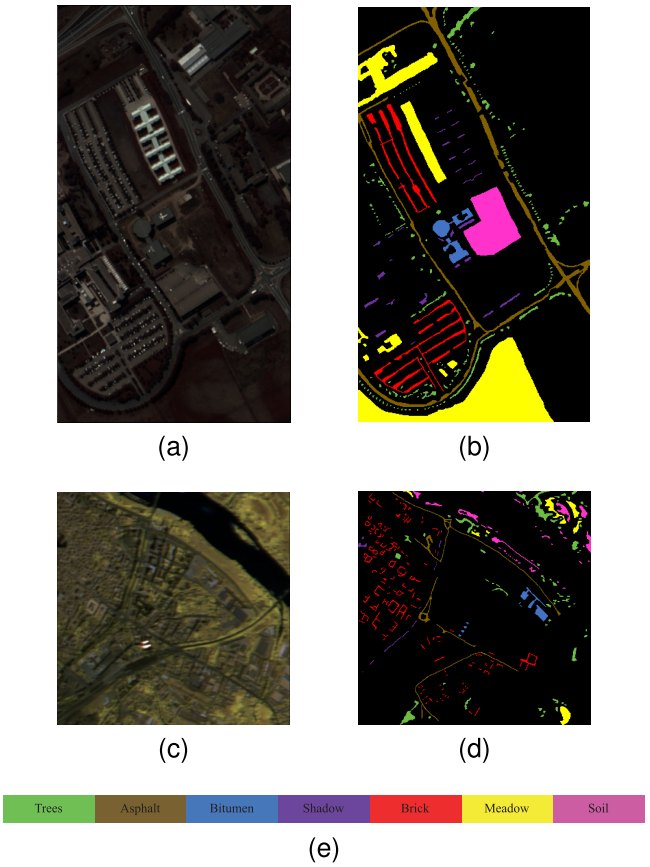

Fig. 7. False-color images and ground-truth maps of RPaviaU-DPaviaC dataset. (a) False-color image of RPaviaU dataset. (b) Ground-truth map of RPaviaU dataset. (c) False-color image of DPaviaC dataset. (d) Ground-truth map of DPaviaC dataset. (e) Class legend.

as the deep learning platform. In the context of cross-domain HSI classification, the source domain offers an abundance of labeled training samples, whereas the target domain consists of a limited number of labeled samples. We randomly select

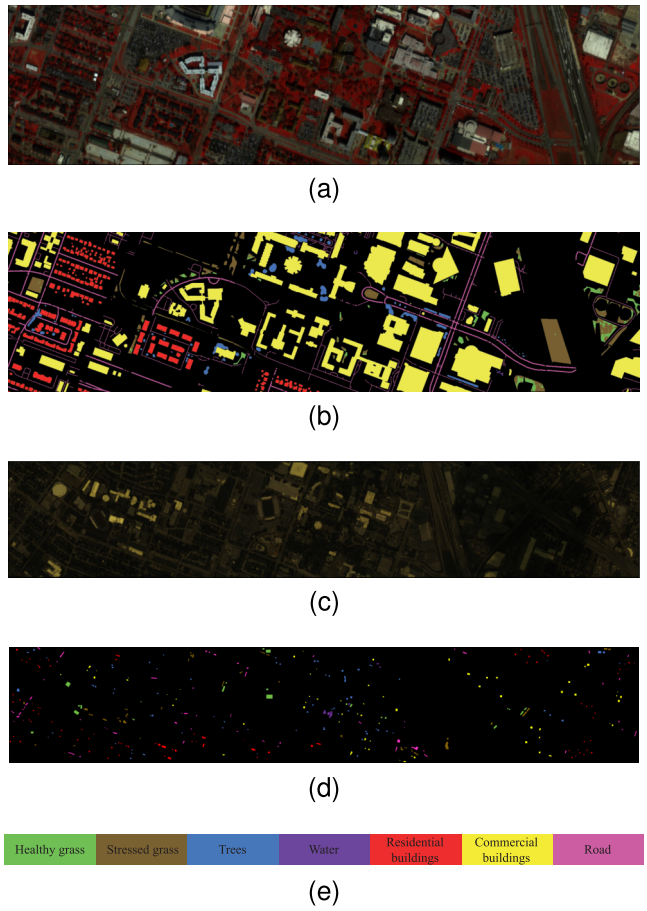

Fig. 8. False-color images and ground-truth maps of Houston2018–Houston2013 dataset. (a) False-color image of Houston2018 dataset. (b) Ground-truth map of Houston2018 dataset. (c) False-color image of Houston2013 dataset. (d) Ground-truth map of Houston2013 dataset. (e) Class legend.

150 labeled source samples and five labeled target samples from each class for training to meet this condition. The remaining target samples constitute the test set.

In order to verify the effectiveness of the CDMC proposed in this article, support vector machine (SVM), hybrid spectral CNN (HybridSN) [56], 3-D separable ResNet (3-D-SRNet) [10], graph information aggregation cross-domain few-shot learning (Gia-CFSL) [19], deep cross-domain few-shot learning (DCFSL) [45], and adaptive domain-adversarial few-shot learning (ADAFSL) [57] are used for comparison. SVM is a traditional machine learning method, and the spectral classifier based on SVM was regarded as the most advanced classification method for single-domain HSI classification at one time [58]. HybridSN is a single-domain HSI classification method based on deep learning. 3-D-SRNet is a fine-tuning-based deep cross-domain HSI classification method. Combining GOT and MMD, Gia-CFSL learns a shared feature space that reduces the impact of domain shifts on cross-domain classification tasks. DCFSL uses adversarial domain adaptation strategies to reduce domain shifts, and thus, domain distribution alignment can be achieved. ADAFSL [57] can assign more focus to classes that exhibit greater discrepancies between the two domains. The settings for each method are shown in Table II. It should be noted that the

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TABLE I
NUMBER OF EACH LAND-COVER CLASS IN EHANGZHOU-RPAVIAHR, RPAVIAU-DPAVIAC, AND HOUSTON2018–HOUSTON2013 DATASETS

No.	EHangzhou-RPaviaHR			RPaviaU-DPaviaC			Houston2018-Houston2013		
	Class	EHangzhou	RPaviaHR	Class	RPaviaU	DPaviaC	Class	Houston2018	Houston2013
1	Water	18043	22525	Trees	3064	2424	Healthy grass	9799	1251
2	Ground/Building	77450	145416	Asphalt	6631	1704	Stressed grass	32502	1254
3	Vegetation	40207	22961	Bitumen	1330	685	Trees	13595	1244
4				Shadow	947	241	Water	266	325
5				Brick	3682	2237	Residential buildings	39772	1268
6				Meadow	18649	1251	Commercial buildings	223752	1244
7				Soil	5029	1475	Road	45866	1252
Total		135700	190902		39332	10017		365552	7838

TABLE II
SETTINGS OF EACH CLASSIFICATION METHOD

Method	Settings
SVM	Kernel: radial basis function (RBF).
	$C \in \{0.001, 0.01, 0.1, \ldots, 100\}$.
	$\gamma \in \{1, 2, 4, \ldots, 32\}$.
HybridSN [56]	Patch size: 25×25.
	Learning rate: 0.001.
	Number of epochs for training:1000.
3-D-SRNet [10]	Patch size: 5×5.
	Learning rate: 0.01.
	Number of epochs for pre-training: 1000.
	Number of epochs for fine-tuning: 1000.
Gia-CFSL [19]	Patch size: 9×9.
	Learning rate: 0.01.
	Number of epochs for training: 1000.
DCFSL [45]	Patch size: 9×9.
	Learning rate: 0.001.
	Number of epochs for training: 1000.
ADAFSL [57]	Patch size: 9×9.
	Learning rate: 0.001.
	Number of epochs for training: 1000.
CDMC	Patch size: 7×7.
	Learning rate: 0.0001.
	Number of epochs for training: 1000.

single-domain and cross-domain classification methods have different experimental settings. The single-domain classification methods, i.e., SVM and HybridSN, are trained with only five labeled samples per class in the target domain. In contrast, the cross-domain classification methods, i.e., 3-D-SRNet, Gia-CFSL, DCFSL, ADAFSL, and our proposed CDMC, are trained with 150 labeled samples per class in the source domain and five labeled samples per class in the target domain.

In our experiments, overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa) are selected as evaluation metrics. The average of ten independent runs is displayed as the experimental results.

C. Discussion on the Length of Chains

The number of cycles, i.e., the value of K, plays a non-negligible role in our proposed CDMC. The value of K determines the length of the CDMC for source and target data. A small value of K leads to insignificant gain from the chains. For instance, when $K = 1$, the model comprises only one cycle, resembling the behavior of a CycleGAN. Conversely, a large value of K leads to much unnecessary time in computation. In order to find the optimal value of K and verify the effect of the parameter, we set $\alpha = 1.0$, $\beta = 1.0$, $\lambda = 1.0$, and assign equal weights of 1.0 to all cycles, i.e., $\gamma_k = 1.0$. In the experiments, we also specifically discuss the role of half-cycles. The experiments on three cross-domain HSI datasets for CDMC are conducted with $K \in \{1, 1.5, 2, \ldots, 6, 6.5, 7\}$. We explain the meaning when K is set to 1, 1.5, 2 in Table III, and the meanings represented by other K values can be inferred in a similar way.

TABLE III
MEANING REPRESENTED BY DIFFERENT K VALUES

K	Source chain	Target chain
1	$\widetilde{\mathbf{f}}_{\mathcal{S}}^{(0)} \rightarrow \widetilde{\mathbf{f}}_{\mathcal{T}}^{(1)} \rightarrow \widetilde{\mathbf{f}}_{\mathcal{S}}^{(1)}$	$\hat{\mathbf{f}}_{\mathcal{T}}^{(0)} \rightarrow \hat{\mathbf{f}}_{\mathcal{S}}^{(1)} \rightarrow \hat{\mathbf{f}}_{\mathcal{T}}^{(1)}$
1.5	$\widetilde{\mathbf{f}}_{\mathcal{S}}^{(0)} \rightarrow \widetilde{\mathbf{f}}_{\mathcal{T}}^{(1)} \rightarrow \widetilde{\mathbf{f}}_{\mathcal{S}}^{(1)} \rightarrow \widetilde{\mathbf{f}}_{\mathcal{T}}^{(2)}$	$\hat{\mathbf{f}}_{\mathcal{T}}^{(0)} \rightarrow \hat{\mathbf{f}}_{\mathcal{S}}^{(1)} \rightarrow \hat{\mathbf{f}}_{\mathcal{T}}^{(1)} \rightarrow \hat{\mathbf{f}}_{\mathcal{S}}^{(2)}$
2	$\widetilde{\mathbf{f}}_{\mathcal{S}}^{(0)} \rightarrow \widetilde{\mathbf{f}}_{\mathcal{T}}^{(1)} \rightarrow \widetilde{\mathbf{f}}_{\mathcal{S}}^{(1)} \rightarrow \widetilde{\mathbf{f}}_{\mathcal{T}}^{(2)} \rightarrow \widetilde{\mathbf{f}}_{\mathcal{S}}^{(2)}$	$\hat{\mathbf{f}}_{\mathcal{T}}^{(0)} \rightarrow \hat{\mathbf{f}}_{\mathcal{S}}^{(1)} \rightarrow \hat{\mathbf{f}}_{\mathcal{T}}^{(1)} \rightarrow \hat{\mathbf{f}}_{\mathcal{S}}^{(2)} \rightarrow \hat{\mathbf{f}}_{\mathcal{T}}^{(2)}$
...	...	...

The classification results obtained by CDMC with different values of K can be seen in Fig. 9. As shown in Fig. 9, when $K < 5$, the evaluation metrics (OA, AA, and Kappa) on EHangzhou-RPaviaHR, RPaviaU-DPaviaC, and Houston2018–Houston2013 datasets generally improve as K increases. When K is greater than 5, we do not discover an apparent trend of improvement in each evaluation metric. We observe that the classification performance may decrease in the half-cycle position (for example, when $K = 5.5$). This may be because there is no constraint from the reconstruction loss in the latter half-cycle, leading to a lack of good feature consistency. Our proposed CDMC performs promising classification on three datasets with $K = 5$. In this article, we set K to 5 as the optimal value.

D. Discussion on the Weight of Each Cycle

Each cycle exerts a distinct influence on the model, making variations in the design of γ_k pivotal for influencing model performance. We set $K = 5$, and formulate four different weighting strategies to determine the optimal approach:

1) **Weighting strategy A**: We adopt a weight of 1.0 only to the last cycle, i.e., $\gamma_K = 1.0$, and $\gamma_k = 0$ when $k < K$. In this case, only the error produced by the last cycle out of all the cycles is calculated.
2) **Weighting strategy B**: We use decreasing weights, i.e., $\gamma_k = 0.9^{k-1}$. In this case, the weight for the first cycle (γ_1) is 1.0.
3) **Weighting strategy C**: We employ increasing weights, i.e., $\gamma_k = 0.9^{K-k}$. In this case, the weight for the last cycle (γ_K) is 1.0.
4) **Weighting strategy D**: We uniformly assign a weight of 1.0 to each cycle, i.e., $\gamma_k = 1.0$.

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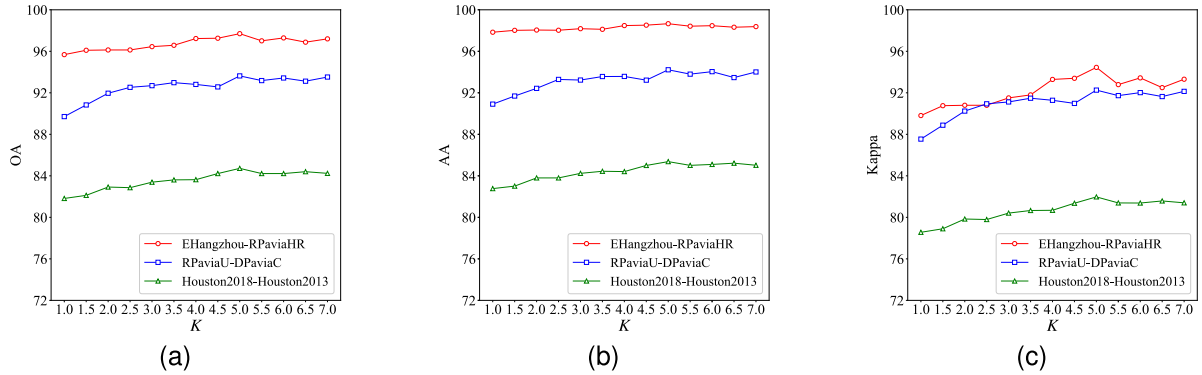

Fig. 9. Classification results obtained by CDMC with different values of K. (a) OA. (b) AA. (c) Kappa.

TABLE IV
DISCUSSION ON WEIGHTING STRATEGIES

Weighting strategy	EHangzhou-RPaviaHR			RPaviaU-DPaviaC			Houston2018-Houston2013		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
A	95.85	97.86	90.16	89.07	89.84	86.75	79.27	80.35	75.59
B	95.98	97.98	90.48	92.78	93.14	91.23	83.67	84.56	80.71
C	96.32	98.14	91.23	93.25	93.84	91.80	83.98	84.86	81.09
D	**97.70**	**98.66**	**94.45**	**93.63**	**94.22**	**92.26**	**84.72**	**85.37**	**81.97**

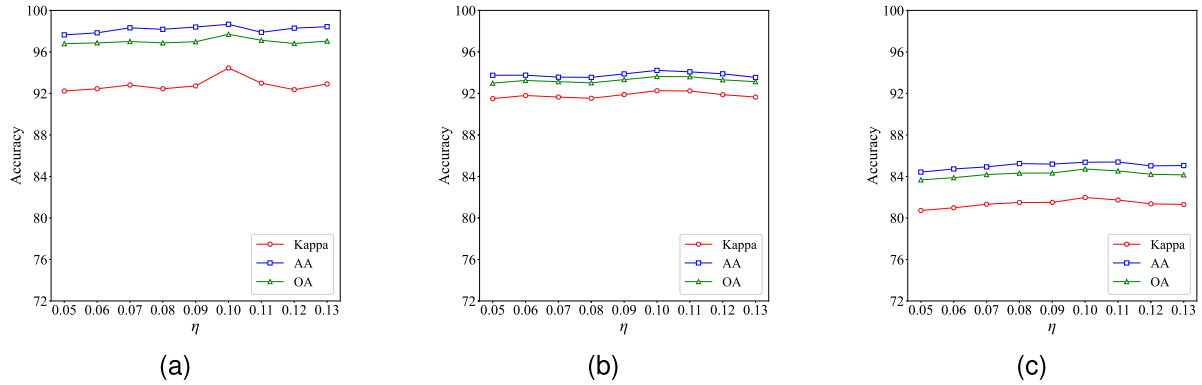

Fig. 10. Classification results obtained by CDMC with different balancing parameters for noise. (a) EHangzhou-RPaviaHR. (b) RPaviaU-DPaviaC. (c) Houston2018–Houston2013.

These distinct weighting strategies enable us to investigate how different cycle weightings affect model performance. The outcomes are presented in Table IV. When employing weighting strategy A, some knowledge is forfeited because the weights of intermediate cycles are set to 0. Consequently, weighting strategy A results in the weakest classification performance. On the other hand, weighting strategies B, C, and D can harness knowledge from intermediate cycles, enhancing the model's performance. In summary, among the four weighting strategies, strategy D is the most advantageous.

E. Discussion on the Intensity of Noise

In this section, we seek an optimal noise intensity. We adopt weighting strategy D for each cycle, with K set to 5. We set $\sigma = \eta A$, where A is the mean of the absolute values of the input features and $\eta \in \{0.05, 0.06, 0.07, \ldots, 0.13\}$. The results on EHangzhou-RPaviaHR, RPaviaU-DPaviaC, and Houston2018–Houston2013, using various values of η, are illustrated in Fig. 10.

TABLE V
OA, AA, AND KAPPA OF EACH METHOD ON EHANGZHOU-RPAVIAHR DATASET

Method	OA	AA	Kappa
SVM	78.94	75.64	64.45
HybridSN	84.42	83.28	65.97
3-D-SRNet	86.97	91.87	72.47
Gia-CFSL	94.65	97.18	87.54
DCFSL	93.61	96.77	85.47
ADAFSL	96.55	98.40	91.91
CDMC	**97.70**	**98.66**	**94.45**

From Fig. 10, it is evident that when $\eta = 0.1$, our proposed CDMC demonstrates satisfactory classification performance across all three datasets. To sum up, we will configure $K = 5$, $\eta = 0.1$, and employ weighting strategy D for the forthcoming experiments.

F. Experimental Results and Analysis

Table V demonstrates the classification results on the EHangzhou-RPaviaHR dataset obtained by each method, and the classification maps are shown in Fig. 11. It should be

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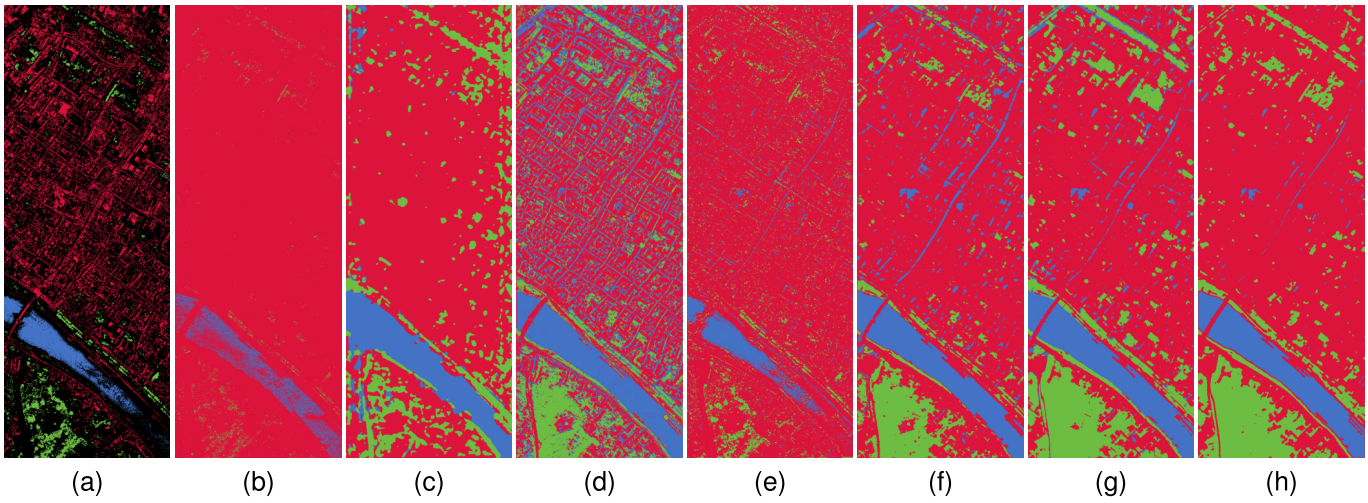

Fig. 11. Classification maps obtained by each method on EHangzhou-RPaviaHR dataset. (a) Ground-truth map. (b) Classification map obtained by SVM. (c) Classification map obtained by HybridSN. (d) Classification map obtained by 3-D-SRNet. (e) Classification map obtained by Gia-CFSL. (f) Classification map obtained by DCFSL. (g) Classification map obtained by ADAFSL. (h) Classification map obtained by our proposed CDMC.

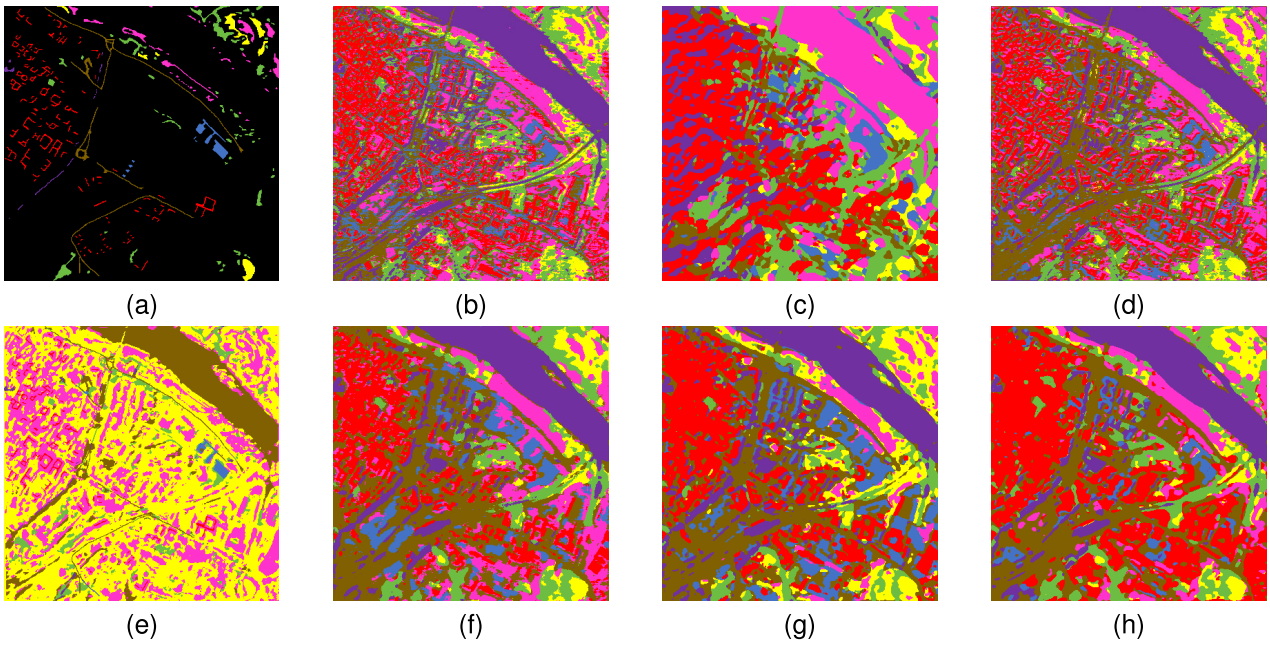

Fig. 12. Classification maps obtained by each method on RPaviaU-DPaviaC dataset. (a) Ground-truth map. (b) Classification map obtained by SVM. (c) Classification map obtained by HybridSN. (d) Classification map obtained by 3-D-SRNet. (e) Classification map obtained by Gia-CFSL. (f) Classification map obtained by DCFSL. (g) Classification map obtained by ADAFSL. (h) Classification map obtained by our proposed CDMC.

TABLE VI

OA, AA, AND KAPPA OF EACH METHOD ON RPAVIAU-DPAVIAC DATASET

Method	OA	AA	Kappa
SVM	82.78	82.06	79.24
HybridSN	71.93	75.17	66.32
3-D-SRNet	88.13	88.96	85.62
Gia-CFSL	87.60	88.66	85.01
DCFSL	89.11	90.24	86.80
ADAFSL	90.94	91.85	89.02
CDMC	**93.63**	**94.22**	**92.26**

TABLE VII

OA, AA, AND KAPPA OF EACH METHOD ON HOUSTON2018–HOUSTON2013 DATASET

Method	OA	AA	Kappa
SVM	65.82	66.71	59.52
HybridSN	64.24	66.77	57.84
3-D-SRNet	72.82	72.85	67.88
Gia-CFSL	77.88	77.86	73.92
DCFSL	79.91	80.16	76.32
ADAFSL	82.19	82.32	78.97
CDMC	**84.72**	**85.37**	**81.97**

noted that the evaluation metrics in the table are calculated only by test samples in the target domain. At the same time, the classification maps in this article are obtained by all pixels of the target HSI, including the background pixels. Through comparison and analysis, the following conclusions are drawn.

1) The cross-domain classification methods (3-D-SRNet, Gia-CFSL, DCFSL, ADAFSL, and CDMC) show better

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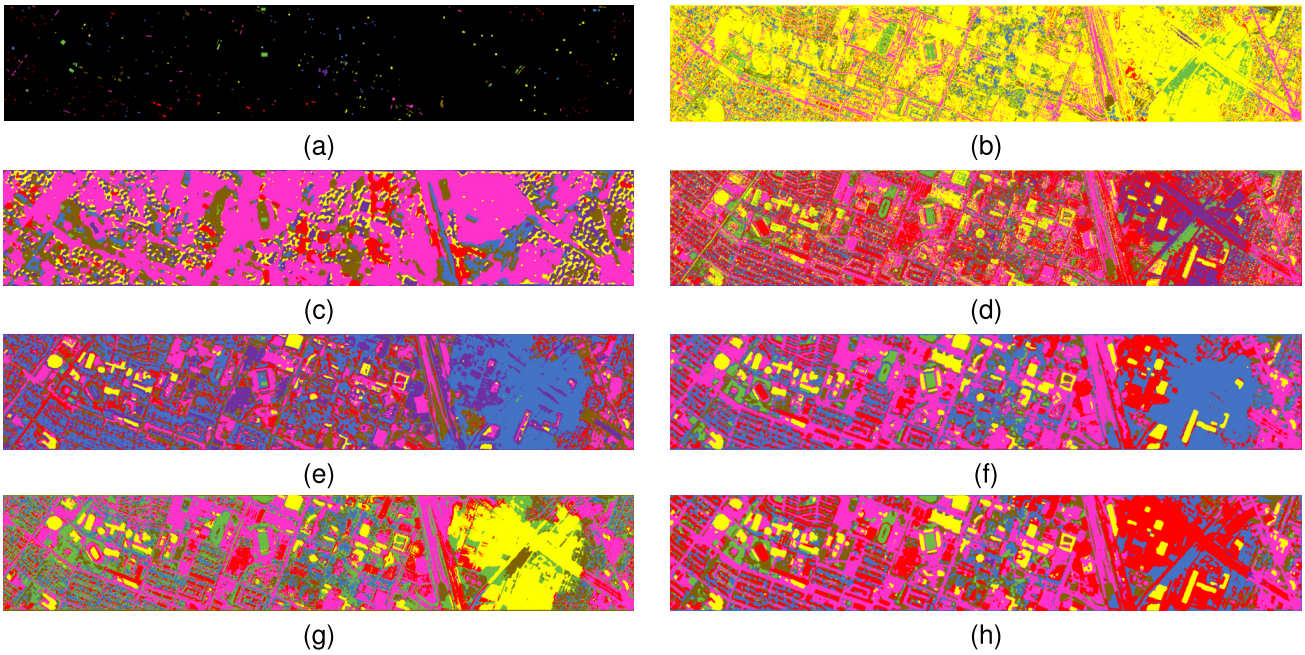

Fig. 13. Classification maps obtained by each method on Houston2018–Houston2013 dataset. (a) Ground-truth map. (b) Classification map obtained by SVM. (c) Classification map obtained by HybridSN. (d) Classification map obtained by 3-D-SRNet. (e) Classification map obtained by Gia-CFSL. (f) Classification map obtained by DCFSL. (g) Classification map obtained by ADAFSL. (h) Classification map obtained by our proposed CDMC.

evaluation metrics overall than single-domain classification methods (SVM and HybridSN) in the cases of the small-sample-size problem, indicating that the knowledge from the source domain can be employed to improve the classification accuracy of the target domain.

2) Compared to other cross-domain classification methods, the CDMC method proposed in this article provides the best classification results, reflecting the advantages of CDMC.

After discussing the experiments on the EHangzhou-RPaviaHR dataset, the results of experiments on the RPaviaU-DPaviaC dataset are analyzed. The classification results on the RPaviaU-DPaviaC dataset obtained by each method are demonstrated in Table VI, and the classification maps are shown in Fig. 12. The following conclusions are drawn.

1) The cross-domain classification methods based on transfer learning remain to show strong competition. Similar to the experimental results on the first EHangzhou-RPaviaHR dataset, the cross-domain classification methods (3-D-SRNet, Gia-CFSL, DCFSL, ADAFSL, and CDMC) outperform the single-domain classification methods (SVM and HybridSN), confirming the effectiveness of transfer learning.
2) As a deep learning-based HSI classification method, HybridSN performs worse than SVM in terms of OA, AA, and Kappa, suggesting that deep learning-based single-domain classification methods may not be suitable for solving small-sample-size problems. Insufficient training samples may make it difficult to train the classifier and, as a result, negatively affect classification performance.
3) Our proposed CDMC is significantly better than other compared cross-domain HSI classification methods in OA, AA, and Kappa, once again highlighting the advantages of CDMC.

The classification results and maps on the Houston2018–Houston2013 dataset obtained by each method are demonstrated in Table VII and Fig. 13, respectively. We can obtain the following conclusions.

1) The cross-domain classification methods still show more advanced classification performance. The evaluation metrics of HybridSN are similar to SVM but significantly lower than those of 3-D-SRNet, Gia-CFSL, DCFSL, ADAFSL, and CDMC. We can obtain that HybridSN cannot solve small-sample-size problems well.
2) It is important to note that our proposed CDMC continues to surpass other compared cross-domain classification methods, demonstrating its high competitiveness in addressing the challenges of heterogeneous cross-domain HSI classification.

G. Ablation Study

We verify the contribution of each strategy in the proposed approach. Each method is described as follows.

1) **CycleGAN**: We remove the auxiliary classifiers from CDMC. It only maps the features in a single cycle, which indicates that the method does not employ the CDMC for source and target data.
2) **Cycle-AC-GAN**: The method includes the auxiliary classifiers and contains a single cycle for source and target data, i.e., $K = 1$. It maps source and target data between domains in one cycle and does not form the CDMC. The comparison between CycleGAN and

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TABLE VIII
OA, AA, AND KAPPA OF EACH METHOD ON EHANGZHOU-RPAVIAHR, RPAVIAU-DPAVIAC, AND HOUSTON2018–HOUSTON2013 DATASETS FOR ABLATION STUDY

Method	Two-way mapping	Auxiliary classifiers	Cross-domain mapping chains	EHangzhou-RPaviaHR			RPaviaU-DPaviaC			Houston2018-Houston2013		
				OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
CycleGAN	✓			95.32	97.73	88.99	87.87	88.80	85.35	78.41	78.49	74.54
Cycle-AC-GAN(K=1)	✓	✓		95.68	97.84	89.81	89.71	90.91	87.54	81.83	82.77	78.56
CDMC(K=5)	✓	✓	✓	**97.70**	**98.66**	**95.45**	**93.63**	**94.22**	**92.26**	**84.72**	**85.37**	**81.97**

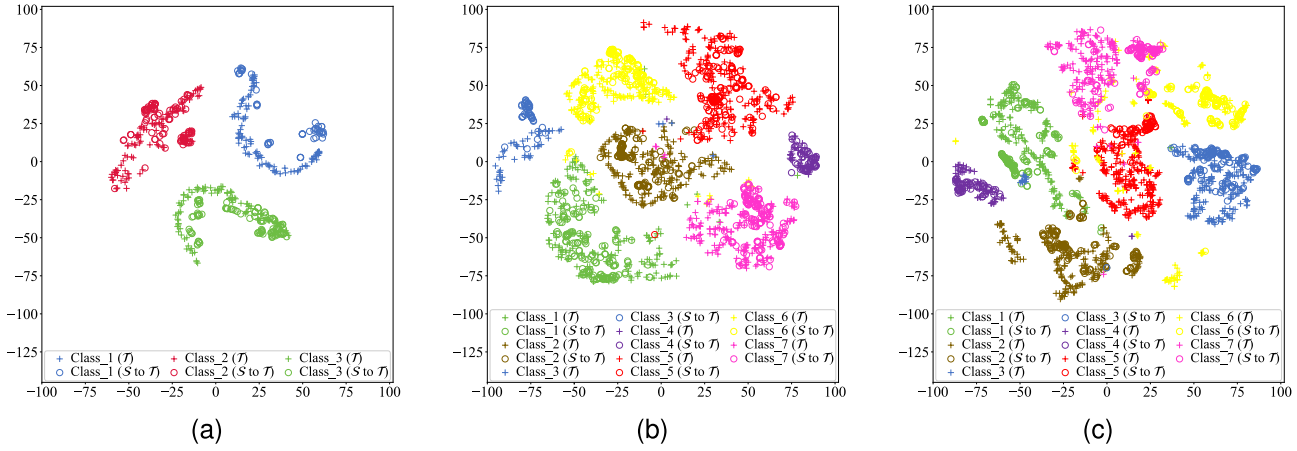

Fig. 14. Visualization of the mapping effects using t-SNE. (a) EHangzhou-RPaviaHR. (b) RPaviaU-DPaviaC. (c) Houston2018–Houston2013.

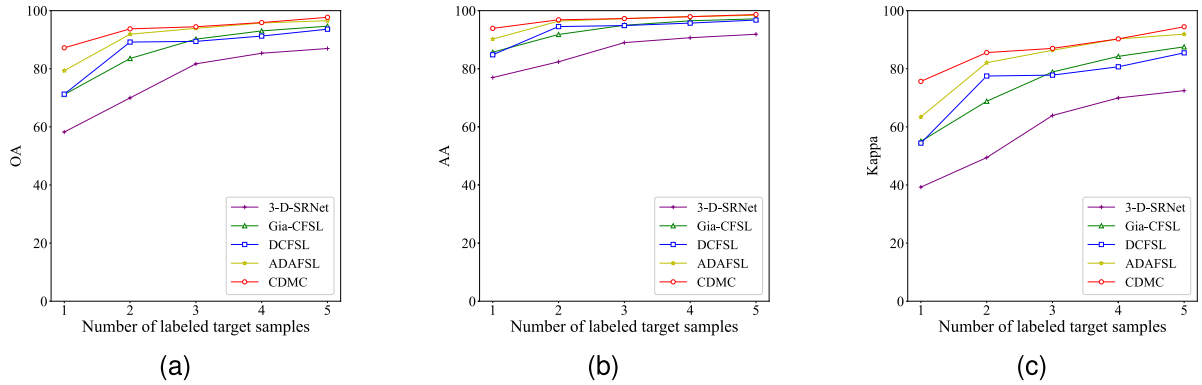

Fig. 15. Classification results with different numbers of labeled target samples on EHangzhou-RPaviaHR dataset. (a) OA. (b) AA. (c) Kappa.

Cycle-AC-GAN will validate the role of the auxiliary classifiers.

3) **CDMC**: CDMC is our proposed method in this article. It contains the auxiliary classifiers and employs the CDMC for source and target data, i.e., $K = 5$. The comparison between Cycle-AC-GAN and CDMC will validate the role of the CDMC.

The classification results for EHangzhou-RPaviaHR, RPaviaU-DPaviaC, and Houston2018–Houston2013 datasets are displayed in Table VIII. The effects of the auxiliary classifiers and CDMC are discussed.

1) **The effects of the auxiliary classifiers**: Cycle-AC-GAN has better classification results than CycleGAN in the three datasets. It illustrates that adding auxiliary classifiers in the mapping benefits classification accuracy. The auxiliary classifiers can aid the consistency of classes in the mapping process.

2) **The effects of the CDMC**: In Table VIII, the classification accuracy of CDMC improves over Cycle-AC-GAN, indicating that the CDMC can assist in the classification of HSIs. Combined with Fig. 9, we can find that a fair number of cycles, i.e., an appropriate length of CDMC, can effectively improve the model's classification performance.

H. Visualization of Mapping and Knowledge Transfer

To demonstrate the effects of mapping and knowledge transfer, we use the t-distributed stochastic neighbor embedding (t-SNE) [59] method to visualize features in a 2-D space. The features processed using t-SNE are shown in Fig. 14, where different classes are shown with different colors, points in the shape of a cross are generated by features of target samples, and points in the shape of a circle are generated by features mapped from the domain $\mathcal{S}$ to the domain $\mathcal{T}$.

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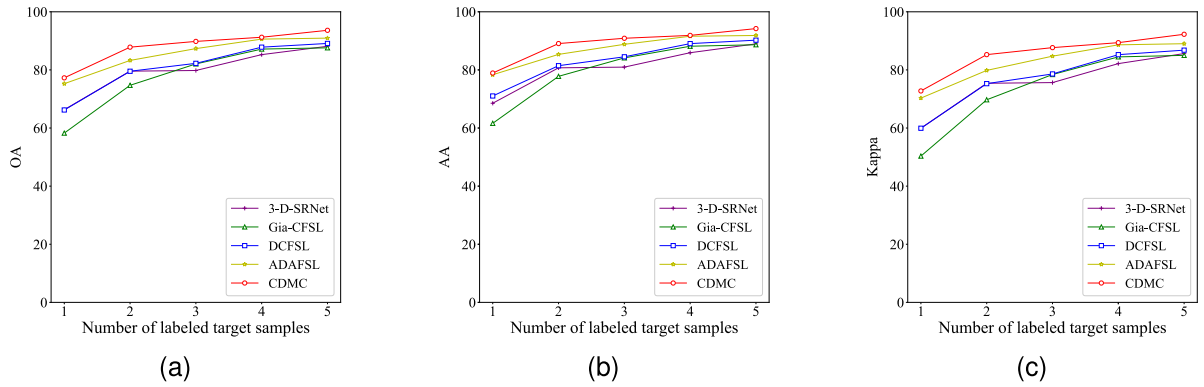

Fig. 16. Classification results with different numbers of labeled target samples on RPaviaU-DPaviaC dataset. (a) OA. (b) AA. (c) Kappa.

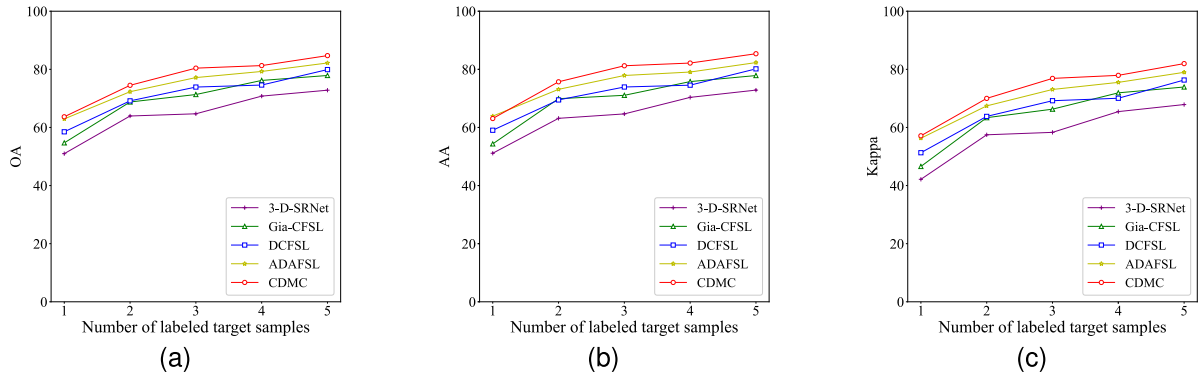

Fig. 17. Classification results with different numbers of labeled target samples on Houston2018–Houston2013 dataset. (a) OA. (b) AA. (c) Kappa.

TABLE IX

OA, AA, AND KAPPA WITH DIFFERENT TRAINING SAMPLE QUALITIES (EHANGZHOU-RPAVIAHR DATASET)

Method	Lower-quality samples			Higher-quality samples		
	OA	AA	Kappa	OA	AA	Kappa
Target-only	89.67	95.29	77.40	99.57	99.37	98.90
CDMC	98.07	98.62	95.18	99.64	99.67	99.09
Improvement	8.40	3.33	17.78	0.07	0.30	0.19

TABLE X

OA, AA, AND KAPPA WITH DIFFERENT TRAINING SAMPLE QUALITIES (RPAVIAU-DPAVIAC DATASET)

Method	Lower-quality samples			Higher-quality samples		
	OA	AA	Kappa	OA	AA	Kappa
Target-only	85.53	87.35	82.58	91.56	93.24	89.80
CDMC	93.20	94.35	91.76	94.25	95.41	93.01
Improvement	7.67	7.00	9.18	2.69	2.17	3.21

TABLE XI

OA, AA, AND KAPPA WITH DIFFERENT TRAINING SAMPLE QUALITIES (HOUSTON2018–HOUSTON2013 DATASET)

Method	Lower-quality samples			Higher-quality samples		
	OA	AA	Kappa	OA	AA	Kappa
Target-only	76.47	77.24	72.17	83.97	83.98	81.01
CDMC	84.17	84.27	81.25	85.71	85.22	83.07
Improvement	7.70	7.03	9.08	1.74	1.24	2.06

We can see that for the same class, the features mapped from the domain $\mathcal{S}$ to the domain $\mathcal{T}$ are mostly close to the features of the target samples, which indicates that our proposed CDMC can align class-conditional distributions during the mapping process.

I. Discussion on the Quantity and Quality of Training Samples

In order to analyze the effects of different numbers of labeled target samples on the performance of cross-domain HSI classification methods (3-D-SRNet, Gia-CFSL, DCFSL, ADAFSL, and CDMC), we select the number of target domain labeled samples from $\{1, 2, 3, 4, 5\}$ for each class, and the classification results of the three datasets are shown in Figs. 15–17. We can find that the more the number of labeled target samples for training, the better the classification performance of the models. However, our proposed CDMC achieves higher classification accuracy even with extremely few training samples compared to other cross-domain HSI classification methods, showing the competitiveness of our proposed CDMC.

We also discuss the effects of the quality of the training samples on the classification results. Specifically, we select the lower-quality samples and higher-quality samples based on the classification results of the Target-only method. Then, the lower and higher-quality samples are used to train our proposed CDMC, respectively. Based on the experimental results of CDMC, the effects of the quality of the training samples on the classification results can be obtained. The experimental results of the three datasets are shown in Tables IX–XI. Our proposed model is more likely to benefit

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TABLE XII
TRAINING TIME (S) OF DIFFERENT DEEP LEARNING-BASED METHODS

Dataset	HybridSN	3-D-SRNet	Gia-CFSL	DCFSL	ADAFSL	CDMC
EHangzhou-RPaviaHR	30.7	267.4	221.3	151.2	140.5	326.2
RPaviaU-DPaviaC	45.4	744.4	363.0	237.0	224.9	693.2
Houston2018-Houston2013	46.0	586.0	368.8	203.6	182.1	635.6

from lower-quality samples and achieve more significant performance improvements.

J. Comparison on Training Time

We record the training time of various deep learning-based methods, including HybridSN, 3-D-SRNet, Gia-CFSL, DCFSL, ADAFSL, and our proposed CDMC, during the experiments. Table XII presents the time cost for these methods on the EHangzhou-RPaviaHR, RPaviaU-DPaviaC, and Houston2018–Houston2013 datasets. It is evident that the training time for cross-domain classification methods (3-D-SRNet, Gia-CFSL, DCFSL, ADAFSL, and CDMC) is significantly longer compared to the single-domain classification method (HybridSN). This discrepancy arises because cross-domain classification methods also require training samples from the domain $\mathcal{S}$. Our proposed CDMC, which is based on adversarial learning and incorporates multiple cycles, has a longer training time, but it remains within an acceptable range.

V. CONCLUSION AND FUTURE WORK

We address the heterogeneous cross-domain HSI classification issue in this article and introduce a novel method, CDMC, to tackle it effectively. The proposed CDMC employs an adversarial learning-based two-way mapping to facilitate knowledge transfer between source and target domains, requiring no paired source-target instances. Notably, this mapping accounts for class-conditional distributions by incorporating auxiliary classifiers. After that, we form multi-CycleGAN by mapping source and target data between domains in multiple cycles. Then, the multi-CycleGAN is unfolded through cycles to build CDMC. The generators in our model provide accurate mappings between domains. Additionally, we accumulate errors throughout the cycles, and backpropagation of these errors ensures that our model effectively harnesses knowledge from each cycle. We conduct experiments on three heterogeneous cross-domain HSI datasets. The experimental results show that through building the CDMC, our proposed CDMC possesses stronger transfer learning performance than other methods.

However, our proposed method mainly uses CNNs, which allows the model to focus only on local information. Currently, several works in the field of remote sensing utilize Transformer to learn a global representation [60], [61]. In the future, we can employ Transformer to extract more comprehensive features for cross-domain HSI classification.

REFERENCES

[1] X. Ma, S. Ji, J. Wang, X. Liu, and H. Wang, "Classification of hyperspectral image based on task-specific learning network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8646–8656, Oct. 2021.

[2] M. Wang, Q. Wang, J. Chanussot, and D. Li, "Hyperspectral image mixed noise removal based on multidirectional low-rank modeling and spatial–spectral total variation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 488–507, Jan. 2021.

[3] M. Wang, Q. Wang, D. Hong, S. K. Roy, and J. Chanussot, "Learning tensor low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 679–691, Jan. 2023.

[4] S. Yu, S. Jia, and C. Xu, "Convolutional neural networks for hyperspectral image classification," *Neurocomputing*, vol. 219, pp. 88–98, Jan. 2017.

[5] K. Gao, W. Guo, X. Yu, B. Liu, A. Yu, and X. Wei, "Deep induction network for small samples classification of hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3462–3477, 2020.

[6] S. Dong, W. Feng, Y. Quan, G. Dauphin, L. Gao, and M. Xing, "Deep ensemble CNN method based on sample expansion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5531815.

[7] Z. Ding and Y. Fu, "Robust transfer metric learning for image classification," *IEEE Trans. Image Process.*, vol. 26, no. 2, pp. 660–670, Feb. 2017.

[8] S. Wang, L. Zhang, W. Zuo, and B. Zhang, "Class-specific reconstruction transfer learning for visual recognition across domains," *IEEE Trans. Image Process.*, vol. 29, pp. 2424–2438, 2020.

[9] Z. Liu, L. Ma, and Q. Du, "Class-wise distribution adaptation for unsupervised classification of hyperspectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 508–521, Jan. 2021.

[10] Y. Jiang, Y. Li, and H. Zhang, "Hyperspectral image classification based on 3-D separable ResNet and transfer learning," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 12, pp. 1949–1953, Dec. 2019.

[11] X. He, Y. Chen, and P. Ghamisi, "Heterogeneous transfer learning for hyperspectral image classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3246–3263, May 2020.

[12] Y. Wang et al., "Heterogeneous few-shot learning for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[13] H. Zhang, Y. Li, Y. Jiang, P. Wang, Q. Shen, and C. Shen, "Hyperspectral classification based on lightweight 3-D-CNN with transfer learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5813–5828, Aug. 2019.

[14] K. Gao, A. Yu, X. You, C. Qiu, B. Liu, and W. Guo, "Learning general-purpose representations for cross-domain hyperspectral images classification with small samples," *Remote Sens.*, vol. 15, no. 4, p. 1080, Feb. 2023.

[15] C. Zhong, J. Zhang, S. Wu, and Y. Zhang, "Cross-scene deep transfer learning with spectral feature adaptation for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 2861–2873, 2020.

[16] C. Deng, Y. Xue, X. Liu, C. Li, and D. Tao, "Active transfer learning network: A unified deep joint spectral–spatial feature learning model for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1741–1754, Mar. 2019.

[17] Y. Su et al., "ACGT-Net: Adaptive cuckoo refinement-based graph transfer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5521314.

[18] M. Hermann, S. Saha, and X. X. Zhu, "Filtering specialized change in a few-shot setting," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1185–1196, 2023.

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[19] Y. Zhang, W. Li, M. Zhang, S. Wang, R. Tao, and Q. Du, "Graph information aggregation cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 2, pp. 1912–1925, Feb. 2024.

[20] W. Li, W. Wei, L. Zhang, C. Wang, and Y. Zhang, "Unsupervised deep domain adaptation for hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2019, pp. 1–4.

[21] Z. Wang, B. Du, Q. Shi, and W. Tu, "Domain adaptation with discriminative distribution and manifold embedding for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 7, pp. 1155–1159, Jul. 2019.

[22] Q. Zhen et al., "Few-shot hyperspectral image classification based on domain adaptation of class balance," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2022, pp. 2255–2258.

[23] Y. Zhang, W. Li, M. Zhang, Y. Qu, R. Tao, and H. Qi, "Topological structure and semantic information transfer network for cross-scene hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 6, pp. 2817–2830, Jun. 2023.

[24] Y. Cheng, W. Zhang, H. Wang, and X. Wang, "Causal meta-transfer learning for cross-domain few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5521014.

[25] Y. Qu, R. K. Baghbaderani, W. Li, L. Gao, Y. Zhang, and H. Qi, "Physically constrained transfer learning through shared abundance space for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10455–10472, Dec. 2021.

[26] S. Li et al., "Discriminative transfer feature and label consistency for cross-domain image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 11, pp. 4842–4856, Nov. 2020.

[27] Y. Wang, Z. Zhang, W. Hao, and C. Song, "Attention guided multiple source and target domain adaptation," *IEEE Trans. Image Process.*, vol. 30, pp. 892–906, 2021.

[28] J. Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 2223–2232.

[29] M. Ye, Y. Qian, J. Zhou, and Y. Y. Tang, "Dictionary learning-based feature-level domain adaptation for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1544–1562, Mar. 2017.

[30] M. B. Bejiga and F. Melgani, "An adversarial approach to cross-sensor hyperspectral data classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 3575–3578.

[31] H. Chen, M. Ye, L. Lei, H. Lu, and Y. Qian, "Semisupervised dual-dictionary learning for heterogeneous transfer learning on cross-scene hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3164–3178, 2020.

[32] S. Jia, S. Jiang, Z. Lin, N. Li, M. Xu, and S. Yu, "A survey: Deep learning for hyperspectral image classification with few labeled samples," *Neurocomputing*, vol. 448, pp. 179–204, Aug. 2021.

[33] C. D. Prakash and L. J. Karam, "It GAN do better: GAN-based detection of objects on images with varying quality," *IEEE Trans. Image Process.*, vol. 30, pp. 9220–9230, 2021.

[34] M. Sultana, A. Mahmood, and S. K. Jung, "Unsupervised moving object detection in complex scenes using adversarial regularizations," *IEEE Trans. Multimedia*, vol. 23, pp. 2005–2018, 2021.

[35] H. Zhang, J. Liu, Z. Yu, and P. Wang, "MASG-GAN: A multi-view attention superpixel-guided generative adversarial network for efficient and simultaneous histopathology image segmentation and classification," *Neurocomputing*, vol. 463, pp. 275–291, Nov. 2021.

[36] Y. Xue et al., "Selective synthetic augmentation with HistoGAN for improved histopathology image classification," *Med. Image Anal.*, vol. 67, Jan. 2021, Art. no. 101816.

[37] L. Ren, L. Gao, M. Wang, X. Sun, and J. Chanussot, "HADGSM: A unified nonconvex framework for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5503415.

[38] B. Li, S. Luo, X. Qin, and L. Pan, "Improving GAN with inverse cumulative distribution function for tabular data synthesis," *Neurocomputing*, vol. 456, pp. 373–383, Oct. 2021.

[39] X. Luo, X. He, X. Chen, L. Qing, and J. Zhang, "DualG-GAN, a dual-channel generator based generative adversarial network for text-to-face synthesis," *Neural Netw.*, vol. 155, pp. 155–167, Nov. 2022.

[40] S. K. Roy, J. M. Haut, M. E. Paoletti, S. R. Dubey, and A. Plaza, "Generative adversarial minority oversampling for spectral–spatial hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5500615.

[41] J. Feng, N. Zhao, R. Shang, X. Zhang, and L. Jiao, "Self-supervised divide-and-conquer generative adversarial network for classification of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536517.

[42] J. Feng, Z. Gao, R. Shang, X. Zhang, and L. Jiao, "Multi-complementary generative adversarial networks with contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5520018.

[43] Y. Ganin et al., "Domain-adversarial training of neural networks," *J. Mach. Learn. Res.*, vol. 17, no. 1, pp. 2030–2096, May 2016.

[44] J. Kuang, G. Xu, T. Tao, and Q. Wu, "Class-imbalance adversarial transfer learning network for cross-domain fault diagnosis with imbalanced data," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–11, 2022.

[45] Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li, and Q. Du, "Deep cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501618.

[46] H. Wang, Y. Cheng, C. L. P. Chen, and X. Wang, "Hyperspectral image classification based on domain adversarial broad adaptation network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517813.

[47] X. Ma, X. Mou, J. Wang, X. Liu, J. Geng, and H. Wang, "Cross-dataset hyperspectral image classification based on adversarial domain adaptation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4179–4190, May 2021.

[48] C. Yu, C. Liu, H. Yu, M. Song, and C.-I. Chang, "Unsupervised domain adaptation with dense-based compaction for hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 12287–12299, 2021.

[49] Z. Zheng, S. Yu, and S. Jiang, "A domain adaptation method for land use classification based on improved HR-Net," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4400911.

[50] X. Wang, Y. Li, and Y. Cheng, "Hyperspectral image classification based on unsupervised heterogeneous domain adaptation CycleGan," *Chin. J. Electron.*, vol. 29, no. 4, pp. 608–614, Jul. 2020.

[51] Y. Liu et al., "CT synthesis from MRI using multi-cycle GAN for head-and-neck radiation therapy," *Computerized Med. Imag. Graph.*, vol. 91, Jul. 2021, Art. no. 101953.

[52] Z. Meng, M. Ye, F. Yao, F. Xiong, and Y. Qian, "Cross-scene hyperspectral image classification based on cycle-consistent adversarial networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 1912–1915.

[53] T. DeVries and G. W. Taylor, "Dataset augmentation in feature space," in *Proc. Int. Conf. Learn. Represent. Workshop*, Apr. 2017, pp. 1–12.

[54] C. Debes et al., "Hyperspectral and LiDAR data fusion: Outcome of the 2013 GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2405–2418, Jun. 2014.

[55] B. Le Saux, N. Yokoya, R. Hansch, and S. Prasad, "2018 IEEE GRSS data fusion contest: Multimodal land use classification [technical committees]," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 1, pp. 52–54, Mar. 2018.

[56] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[57] Z. Ye, J. Wang, H. Liu, Y. Zhang, and W. Li, "Adaptive domain-adversarial few-shot learning for cross-domain hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5532017.

[58] L. Zhu, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5046–5063, Sep. 2018.

[59] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, pp. 2579–2605, Nov. 2008.

[60] M. Wang, B. Zhang, J. Li, L. Ren, and J. Chanussot, "A feature fusion-based transformer network for hyperspectral super-resolution," in *Proc. Workshop Hyperspectral Image Signal Proces., Evol. Remote Sens.*, 2023, pp. 1–5.

[61] M. Bi, M. Wang, Z. Li, and D. Hong, "Vision transformer with contrastive learning for remote sensing image scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 738–749, 2023.

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